

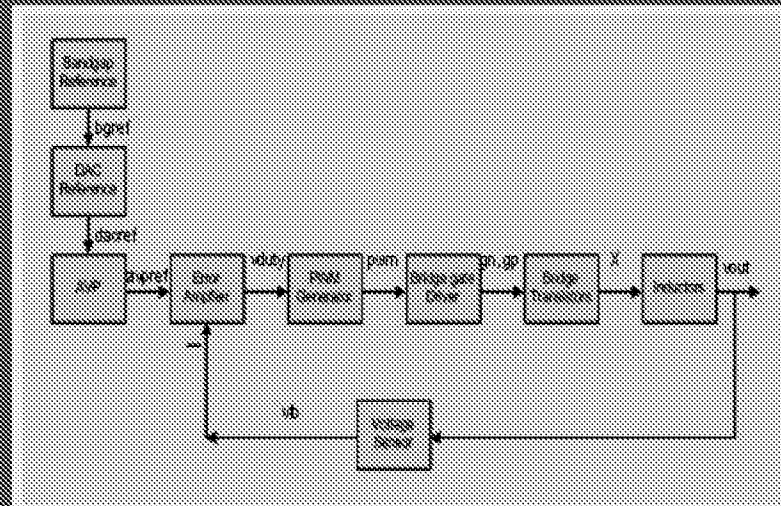
APPENDIX 1

Part 2 of 5

Power Cell Circuits

- VR Loop
- Feedback Control
- PWM & Phase generation
- Bridge driver and Transformer
- Reference
- Bandgap with wideband PSR
- VID control D/A converter
- Linear regulators for noise isolation
- Other features
- Synthetic loads – support test activities
- I, V, T sensors

VR Loop circuits

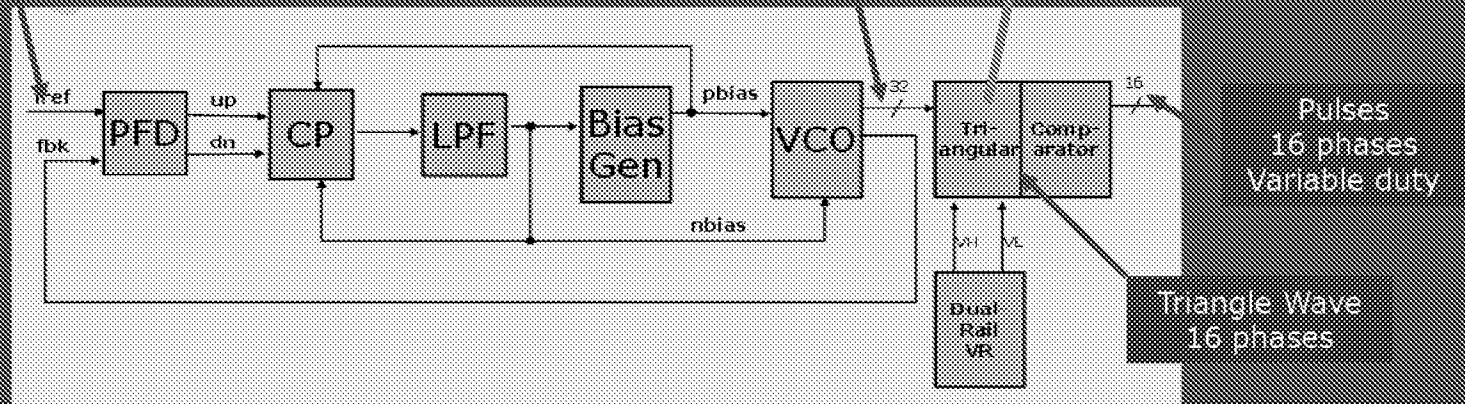
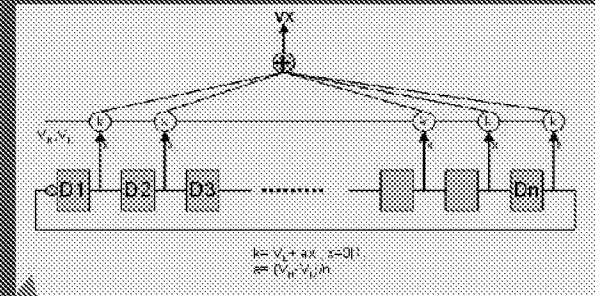


PWM topology

- Differential low power self-biased PLL
- Pulse width control by:
 - Clock => Triangle wave => pulses with variable duty cycle

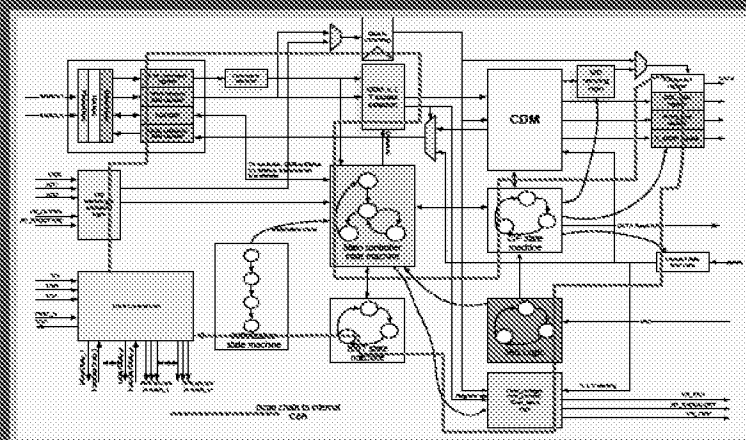
Ref Clock
1 phase
50% duty

Clock
16 phases
50% duty



Master Control Architecture

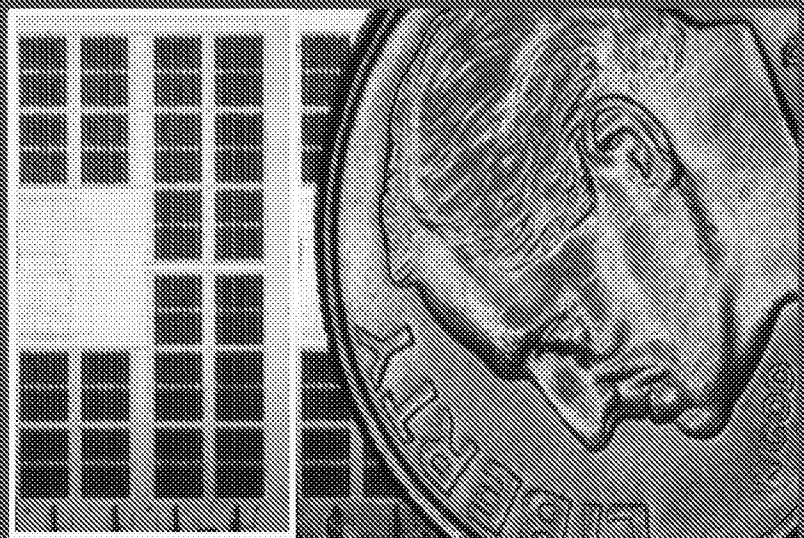
- Master Controller Custom RTL
- VID controller
- JTAG 1194 compliant
- Cell Domain Map (V,T,I)
- AVP adjust
- Test & BIST per cell
- Softstart & Warmstart Algorithms
- Internal Buss interface logic
- IRQ buss
- Platform Interface support
 - Parallel buss
- Cell I-balance



Master Controller Block

Agenda

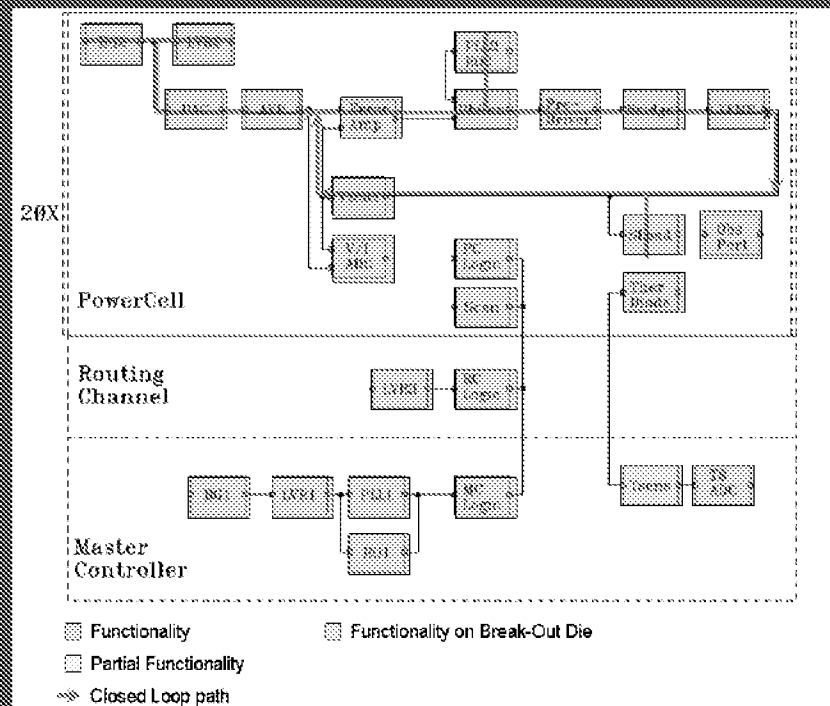
- Architecture
- Circuits
- Results
- Conclusion



Circuit Testing - Schematically

- Validation

- Circuits broken down to analog and digital
- Analog circuits highly observable thru multiple ports
- ADC and DAC conversions for digital readout
- Micro-controller testability thru parallel interface and thru TAP interfaces.



Test Results: Snap-shot of circuits – both Wafer Probe & Package

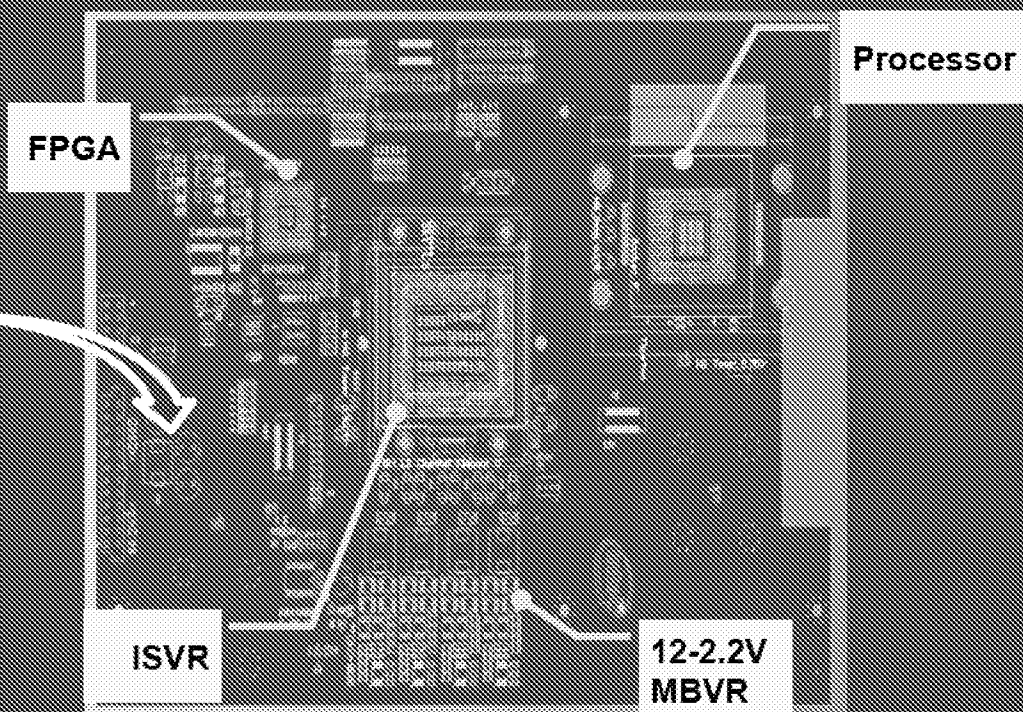
- Bandgap
 - All internal linear regulators (LDO's)
 - Sensors
 - V/I sensors & ADC used for known-good-die screening –all were functional (not fully debugged though)
 - Temperature sensor and its ADC functional on break-out die
 - Interface logic
 - Enabled full programming through either parallel bus or scan
 - DFT features
 - such as manual programming of VCO frequency
 - Observability ports
 - To look at pre-determined internal nodes
- AVP
- Shared all 20 Cells



Platform Testing: Interposer & Microprocessor as load

- For testing with CPU and bench testing

Modified Test Platform
(Used for booting CPU)

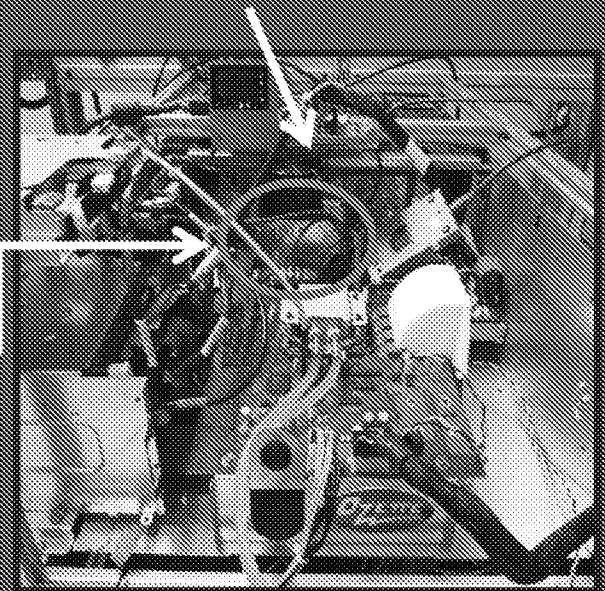
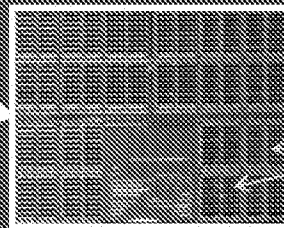
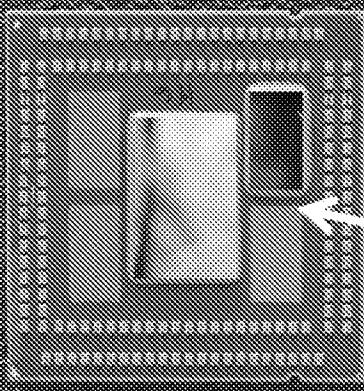


Booted 90W Server CPU – only 3 cells

Intel® Xeon® Processor E7330

- Used only 3 Power Cells
- With 40% of Output Filter Cap*
- Continuous operation with virus for 4+hours
 - No Errors

Ozette



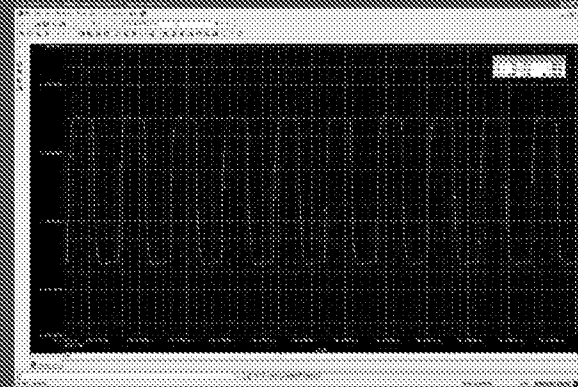
These 3 cells

*Compared with MBVR

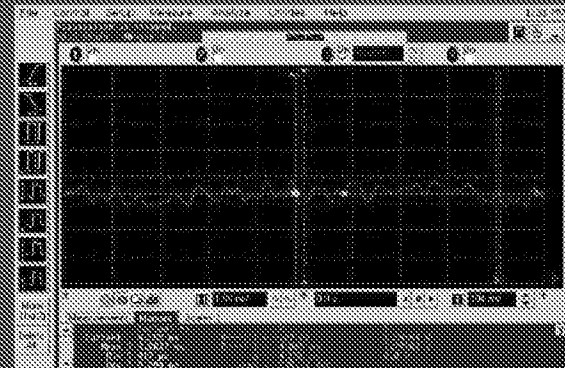


V_{ripple} & V_{TT}

- Voltage Ripple V_{ripple}
- Measurements in lab on ISVR indicate ripple is almost non-existent
- Simulations yielded worst case $\pm 2mV$
- V_{TT} Thermal drift
- Due to thermal time-constant of measurement and error due to linearity circuitry in controller, etc.
- Drift range on package is 60-110C
- Most error is calibrated out and leftover is linearized over temperature range to less than $1mV$



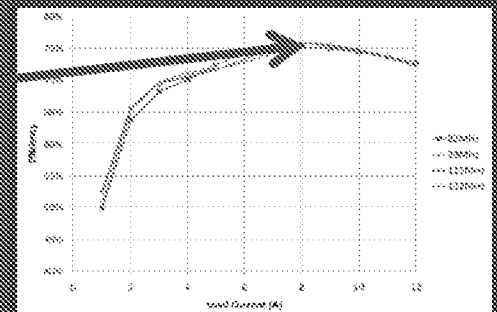
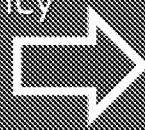
$2mV > V_{ripple}$ sim on Cell



V_{ripple} measurement –
Only noise pickup

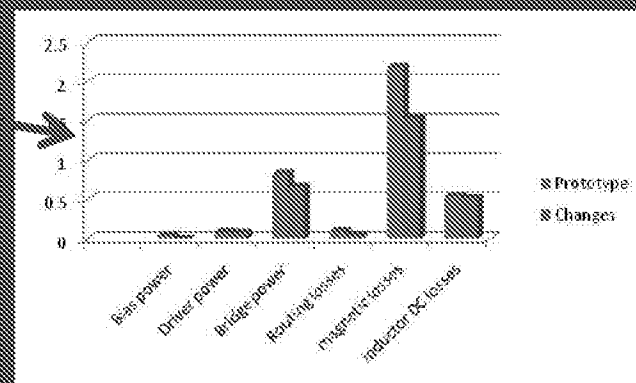
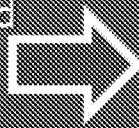
Efficiency – one cell - WIP

- Basic Test (no changes) ~76% peak
 - No bridge shed enable (flat efficiency algorithms not enabled)
 - Bias circuits all on.



'Raw' Efficiency Measurements

- Efficiency* ~82% speculated with basic changes for 'product' level intro
 - Inductor topology coupling change
 - Non-lab level magnetics processed
 - Bias pwr re-distributed
 - Driver/Bridge circuits re-biased
 - Non-test bridge/output routing



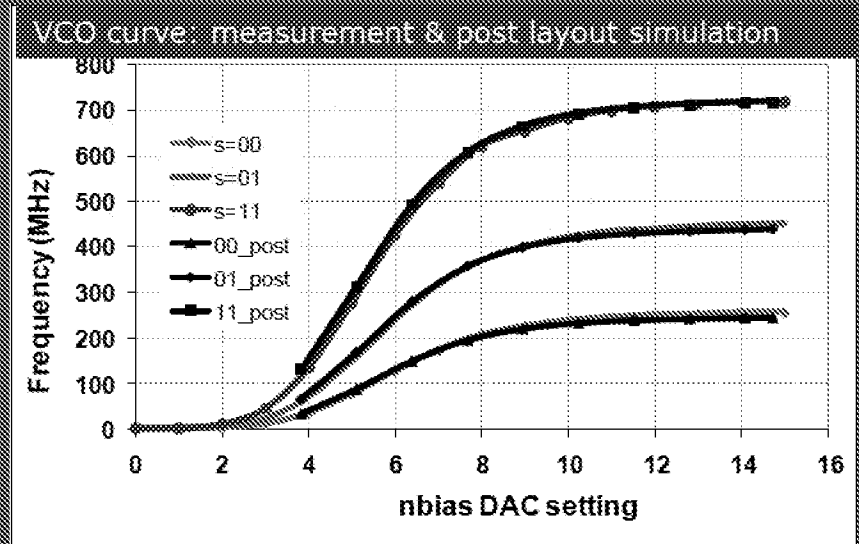
Power Breakdown by element

*Does not include additional advancements that Cannot be reported at this time.



Test Result: Powercell PLL/VCO

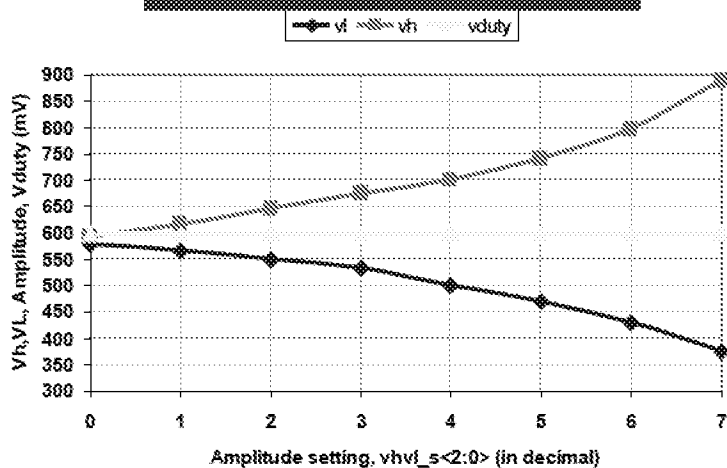
- PLL locking from 20MHz to 200MHz
- Frequencies correlate well with post layout simulation
- Targeted switching frequency, 50-100MHz, is at the linear portion of K_{vco} curve.



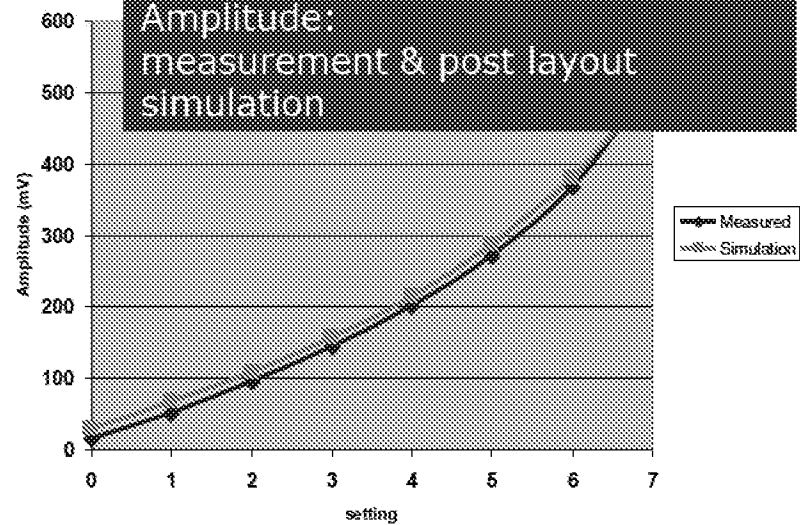
Test Result: Triangle Wave Amplitude

- Triangle Wave amplitude is set by VH/VL control circuit
- A wide range of triangle wave amplitude can be obtained
- Measurement and simulation matched well

VH/VL vs control setting

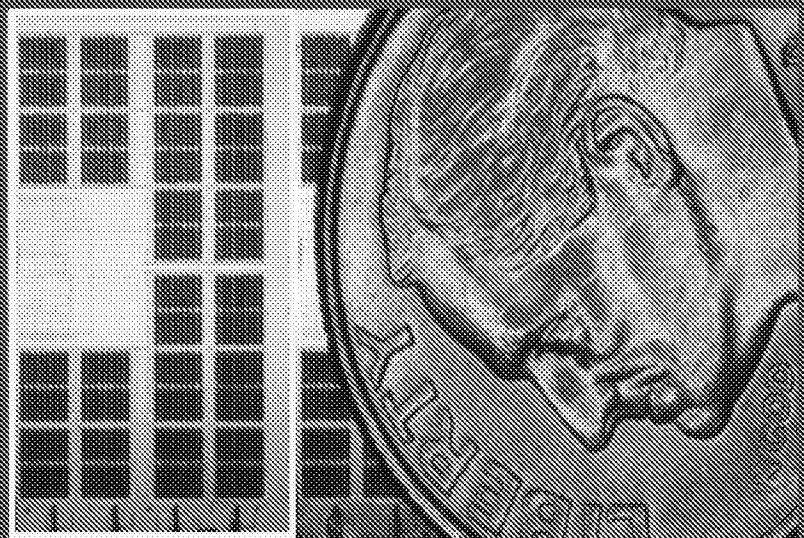


Amplitude:
measurement & post layout
simulation



Agenda

- Architecture
- Circuits
- Results
- Conclusion

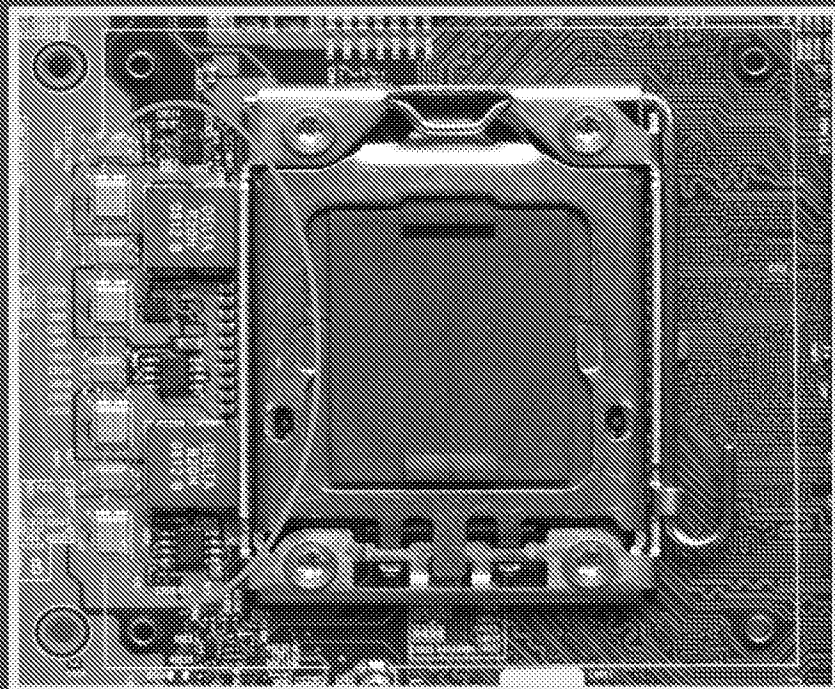


Comparison with other solutions...

- ISVR vs. Platform VR
 - ISVR is $\sim 400A$ design; the other is $\sim 120A$ – input voltages different...but you get the idea.
 - $\sim 110\text{ mm}^2$ vs. 2700 mm^2
- ISVR technology is $\sim 50\times$ smaller


ISVR
 $\sim 12\times 9$
mm

$\sim 18\times 50\text{ mm}$
*1 ISVR +
small 12-
2.4V VR
replaces 3.2
of these!*



VR for Intel® Xeon®
Processor E7330



Conclusions

- 400A capable – tested to 220A for less than ½ of chip
 - Board thermally limited.
- Booted and ran server processor (90W design) with 2 cells – ran with 3 cells under Linpack™ for 4+ hours.
- Ripple below noise threshold
- Efficiency in low 80's with minor changes
 - Additional changes possible will boost up.
- Density is ~8A/mm² thermally constrained
- Questions?





European Patent Office
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Questions about this communication ?

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ROYAUME UNI

Date

10.04.17

Reference JL83874P.EPP	Application No./Patent No. 16173089.0 - 1809 / 3151411
Applicant/Proprietor VLT, Inc.	

Communication pursuant to Rules 70(2) and 70a(2) EPC and reference to Rule 39(1) EPC

The date on which the European Patent Bulletin mentions the publication of the European search report for the above-identified European patent application (publication number: 3151411) is 05.04.17.

Since the request for examination has been filed (R. 70(1), Art. 94(1) EPC) prior to the transmission of the European search report, you are hereby invited to indicate whether you desire to proceed further with the application.

If you do not reply to this invitation within **six months** of the above-mentioned publication date, the application **will be deemed to be withdrawn** (R. 70(3) EPC).

You are invited, within the above-mentioned six-month period, to comment on the objections raised in the opinion accompanying the European search report and/or to file any amendments to the description, claims and drawings correcting any deficiencies noted in the opinion (R. 70a(2), R. 137(2) EPC; Guidelines for Examination in the EPO, B-XI, 8).

If filing amendments, you must identify them and indicate the basis for them in the application as filed. Failure to meet either requirement may lead to a communication from the Examining Division requesting that you correct this deficiency (R. 137(4) EPC).

Should the reply to the invitation pursuant to Rule 70a(2) EPC be filed in an admissible non-EPO language, a translation is to be submitted within one month of its filing (R. 6(2) EPC).

Should you not comply with this invitation within the time limit, the application **will be deemed to be withdrawn** in accordance with Rule 70a(3) EPC.

Note

If not yet paid, the designation fee is also payable within **six months** of the above-mentioned publication date (R. 39(1) EPC).

This period for payment is applicable for extension and validation fees as well.

The present amount of the designation fee is EUR 585,00.

Note to users of the automatic debiting procedure

Unless the EPO receives prior instructions to the contrary, the designation fee will be debited on the last day of the period for payment. For further details see the Arrangements for the automatic debiting procedure, supplementary publication 3 - OJ EPO 2015.

Important information concerning fee amounts

Following any amendment to the Rules relating to Fees, the amount(s) mentioned in this communication may be different from the amount(s) **actually due on the date of payment**. The latest version of the Schedule of fees and expenses, published as a Supplement to the Official Journal of the EPO, is also available on the EPO website (www.epo.org) and can be found under www.epo.org/schedule-of-fees, which allows the viewing, downloading and searching for individual fee amounts, both current and previous.

Please note that procedural fees are usually adjusted every two years, on even years, with effect from 1 April.

Payments by cheque delivered or sent direct to the EPO are no longer accepted as from 1 April 2008 (see OJ EPO 2007, 626).

Note concerning renewal fees

Under Article 86 EPC renewal fees shall be paid to the EPO for the European patent application. These fees shall be due in respect of the third and each subsequent year, calculated from the date of filing. Due account should be taken that the EPO will not send invoices or reminders to pay any fees in due time.

Receiving Section



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Questions about this communication ?

Contact Customer Services at www.epo.org/contact



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ROYAUME UNI

Date

14.02.17

Reference JL83874P.EPP	Application No./Patent No. 16173089.0 - 1809
Applicant/Proprietor VLT, Inc.	

Communication

The extended European search report is enclosed.

The extended European search report includes, pursuant to Rule 62 EPC, the European search report (R. 61 EPC) or the partial European search report/ declaration of no search (R. 63 EPC) and the European search opinion.

Copies of documents cited in the European search report are attached.

☒ 0 additional set(s) of copies of such documents is (are) enclosed as well.

The following have been approved:

☒ Abstract ☒ Title

☐ The Abstract was modified and the definitive text is attached to this communication.

The following figure(s) will be published together with the abstract: 1

Refund of search fee

If applicable under Article 9 Rules relating to fees, a separate communication from the Receiving Section on the refund of the search fee will be sent later.

Should you wish to further prosecute this application in the examination phase, your attention is drawn to the provisions of Rule 70a EPC. An invitation to respond to the extended European search report will be issued once the date of publication of the European search report has been mentioned in the European Patent Bulletin (R. 69(1), R. 70(2) EPC).



The examination is being carried out on the **following application documents**

Description, Pages

1-8 as originally filed

Claims, Numbers

1-15 as originally filed

Drawings, Sheets

1/3-3/3 as originally filed

1 Reference is made to the following documents; the numbering will be adhered to in the rest of the procedure.

D1 JP H02 280666 A (SANKEN ELECTRIC CO LTD) 16 November 1990 (1990-11-16)

D2 EP 2 863 531 A1 (PANASONIC IP MAN CO LTD [JP]) 22 April 2015 (2015-04-22)

Conciseness

2 The application does not meet the requirements of Article 84 EPC, because claims 1, 8, 10, 11 and 12 are not concise.

2.1 Claims 1, 8, 10, 11 and 12 have been drafted as separate independent claims.

Under Article 84 in combination with Rule 43(2) EPC, an application may contain more than one independent claim in a particular category only if the subject-matter claimed falls within one or more of the exceptional situations set out in paragraph (a), (b) or (c) of Rule 43(2) EPC. This is not the case in the present application, however, the claims all appearing to relate to the same method of converting power.

- 2.2 In the further prosecution of the application, failure to file an amended set of claims which complies with Rule 43(2) EPC, or to submit convincing arguments as to why the current set of claims does in fact comply with these provisions, may lead to refusal of the application under Article 97(2) EPC.

Novelty

- 3 The present application does not meet the requirements of Article 52(1) EPC because the subject-matter of claims 1, 2, 8, 10, 11 and 12 is not new within the meaning of Article 54(1) and (2) EPC.

- 3.1 Document D1 discloses:

A power conversion apparatus for converting power between an input source and a load (abstract), where the load may vary over a normal operating range (abstract, last sentence), comprising:

a transformer (figure 1:T);

a resonant circuit including the transformer and a resonant capacitor and having a characteristic resonant frequency and period (figure 1:L₁,C₁ and abstract);

output circuitry (figure 1:7) connected to the transformer for delivering a rectified output voltage to the load;

two or more primary switches (figure 1:Q1,Q2) to drive the resonant circuit;

a clamp switch (figure 1:S1,S2) to clamp the resonant capacitor; and

a switch controller (figure 1:25) to operate the primary switches and the clamp switch in a series of converter operating cycles, each converter operating cycle comprising:

(a) two power transfer intervals (figure 2:periods t₀-t₆) of essentially equal duration during which one or more of the primary switches are ON, a magnetizing current flows in a winding of the transformer, and power is transferred between the input and the output via the transformer, each power transfer interval comprising:

(i) first and second resonant intervals (figure 2:t2-t4 and corresponding period in second half-wave), each having a duration less than the characteristic resonant period, during which a resonant current flows through a winding of the transformer at the characteristic resonant frequency; and

(ii) a clamp interval (figure 2:S2 high period) occurring after the first resonant interval and before the second resonant interval and having a clamp duration during which the clamp switch is ON and provides a low impedance shunt across the resonant capacitor.

- 3.2 The definition of the converter operating cycles is open in claim 2 and may therefore correspond to two switching periods, each corresponding to a power transfer interval. The subject-matter of claim 1 is therefore not new.
- 3.3 The subject-matter of independent claim 1 corresponds in terms of method features to that of claim 2. The objections raised with respect to the latter claim, therefore, also apply, mutatis mutandis, to independent claim 1, which is therefore not allowable under Article 52(1) EPC.
- 3.4 The subject-matter of claims 8, 10, 11 and 12 corresponds to that of claim 1. The objections raised with respect to the latter claim, therefore, also apply, mutatis mutandis, to independent claims 8, 10, 11 and 12, which are therefore not allowable under Article 52(1) EPC.

Further Remarks

- 4 The invention as currently claimed includes in its scope embodiments in which the transformer and resonant capacitor are connected in parallel. Such embodiments do not, however, appear to be supported by the description (Article 84 EPC), which does not supply the skilled reader with sufficient information to allow him to adjust the durations of the output current pulses for supplying the varying load with such a configuration.

Amendments

5 The applicant is invited to file new claims which take account of the above comments.

5.1 To meet the requirements of Rule 42(1)(b) EPC, document D2 should be identified in the description and its relevant contents should be indicated. The applicant should ensure that it is clear from the description which features of the subject-matter of independent claims are known from the document D2.

5.2 Independent claims should be in the two-part form in accordance with Rule 43(1) EPC, with those features known in combination from the prior art being placed in the preamble (Rule 43(1)(a) EPC) and the remaining features being included in the characterising part (Rule 43(1)(b) EPC).

5.3 The features of the claims should be provided with reference signs placed in parentheses to increase the intelligibility of the claims (Rule 43(7) EPC). This applies to both the preamble and characterising portion (see Guidelines F-IV, 4.19).

5.4 When filing amended claims the applicant should at the same time bring the description into conformity with the amended claims. Care should be taken during revision, especially of the introductory portion and of any statements of problem or advantage, not to add subject-matter which extends beyond the content of the application as originally filed (Article 123(2) EPC).

5.5 In order to comply with the requirements of Rule 137(4) EPC, the applicant should clearly identify the amendments made, irrespective of whether they concern amendments by addition, replacement or deletion, and indicate the passages of the application as filed on which these amendments are based (see Guidelines H III, 2.1).

If the applicant considers it appropriate, these indications could be submitted in handwritten form on a copy of the relevant parts of the application as filed.

Information on Search Strategy - Pilot phase (see OJ 2015, A86)

The type of information contained in this sheet may change during the pilot for improving the usefulness of this new service.

Application Number

EP 16 17 3089

TITLE: CLAMPED CAPACITOR RESONANT POWER CONVERTER

APPLICANT: VLT, Inc.

IPC CLASSIFICATION: H02M3/337, H02M1/00

EXAMINER: Lochhead, Steven

CONSULTED DATABASES: NPL, WPI

CLASSIFICATION SYMBOLS DEFINING EXTENT OF THE SEARCH:

IPC:

CPC: H02M3/337, H02M3/3376, H02M2001/0058, H02M3/28/LOW, H02M2007/4818, 5H730/BB27, 5H730/BB66, 5H730/FG21, H02M3/33569, H02M2007/4815, H02M2005/2937, 5H750/DD06, H01R4/646

FI/F-TERMS:

KEYWORDS OR OTHER ELEMENTS FEATURING THE INVENTION:

full-bridge, resonant, quasi-resonant, shunt, clamp, parallel, switch

EUROPEAN SEARCH REPORT

Application Number
EP 16 17 3089

DOCUMENTS CONSIDERED TO BE RELEVANT			
Category	Citation of document with indication, where appropriate, of relevant passages	Relevant to claim	CLASSIFICATION OF THE APPLICATION (IPC)
X	JP H02 280666 A (SANKEN ELECTRIC CO LTD) 16 November 1990 (1990-11-16)	1,8, 10-12	INV. H02M3/337
A	* abstract * * figures 1,2 *	2-7,9, 13-15	ADD. H02M1/00
A	----- EP 2 863 531 A1 (PANASONIC IP MAN CO LTD [JP]) 22 April 2015 (2015-04-22) * the whole document *	1-15	
A	----- Shizumasa Okudaira ET AL: "A New Quasi-Resonant Inverter With Two-way Short-circuit Switch Across A Resonant Capacitor", Proceedings of the Power Conversion Conference, 2002. PCC-Osaka 2002., 5 April 2002 (2002-04-05), XP055340750, DOI: 10.1109/PCC.2002.998195 Retrieved from the Internet: URL:http://ieeexplore.ieee.org/ielx5/7830/21539/00998195.pdf?tp=&arnumber=998195&isnumber=21539 [retrieved on 2017-01-31] * the whole document *	1-15	
			TECHNICAL FIELDS SEARCHED (IPC)
			H02M H01R
The present search report has been drawn up for all claims			
Place of search The Hague		Date of completion of the search 7 February 2017	Examiner Lochhead, Steven
CATEGORY OF CITED DOCUMENTS X : particularly relevant if taken alone Y : particularly relevant if combined with another document of the same category A : technological background O : non-written disclosure P : intermediate document T : theory or principle underlying the invention E : earlier patent document, but published on, or after the filing date D : document cited in the application L : document cited for other reasons & : member of the same patent family, corresponding document			

2

**ANNEX TO THE EUROPEAN SEARCH REPORT
ON EUROPEAN PATENT APPLICATION NO.**

EP 16 17 3089

This annex lists the patent family members relating to the patent documents cited in the above-mentioned European search report.
The members are as contained in the European Patent Office EDP file on
The European Patent Office is in no way liable for these particulars which are merely given for the purpose of information.

07-02-2017

Patent document cited in search report	Publication date	Patent family member(s)	Publication date
JP H02280666 A	16-11-1990	JP H0746903 B2	17-05-1995
		JP H02280666 A	16-11-1990

EP 2863531 A1	22-04-2015	EP 2863531 A1	22-04-2015
		JP 5906418 B2	20-04-2016
		JP 2014003764 A	09-01-2014
		US 2015124490 A1	07-05-2015
		WO 2013186991 A1	19-12-2013

Dual Synchronous Rectified MOSFET Drivers

The ISL6210 integrates two ISL6208A drivers and is optimized to drive two independent power channels in a synchronous-rectified buck converter topology. These drivers combined with an Intersil ISL62xx multiphase PWM controller forms a complete single-stage core-voltage regulator solution with high efficiency performance at high switching frequency for advanced microprocessors.

The IC is biased by a single low voltage supply (5V), minimizing driver switching losses in high MOSFET gate capacitance and high switching frequency applications. Each driver is capable of driving a 3nF load with less than 10ns rise/fall time. Bootstrapping of the upper gate driver is implemented via an internal low forward drop diode, reducing implementation cost, complexity, and allowing the use of higher performance, cost effective N-Channel MOSFETs. Adaptive shoot-through protection is integrated to prevent both MOSFETs from conducting simultaneously.

The ISL6210 features 4A typical sink current for the lower gate driver, enhancing the lower MOSFET gate hold-down capability during PHASE node rising edge, preventing power loss caused by the self turn-on of the lower MOSFET due to the high dV/dt of the switching node.

The ISL6210 also features an input that recognizes a high impedance state, working together with Intersil multiphase PWM controllers to prevent negative transients on the controlled output voltage when operation is suspended. This feature eliminates the need for the Schottky diode that may be utilized in a power system to protect the load from negative output voltage damage.

Ordering Information

PARTNUMBER (Note)	PART MARKING	TEMP. RANGE (°C)	PACKAGE (Pb-Free)	PKG. DWG. #
ISL6210CRZ	62 10CRZ	-10 to +100	16 Ld 4x4 QFN	L16.4x4
ISL6210CRZ-T*	62 10CRZ	-10 to +100	16 Ld 4x4 QFN	L16.4x4

*Please refer to TB347 for details on reel specifications.

NOTE: These Intersil Pb-free plastic packaged products employ special Pb-free material sets, molding compounds/die attach materials, and 100% matte tin plate plus anneal (e3 termination finish, which is RoHS compliant and compatible with both SnPb and Pb-free soldering operations). Intersil Pb-free products are MSL classified at Pb-free peak reflow temperatures that meet or exceed the Pb-free requirements of IPC/JEDEC J STD-020.

Features

- 5V Quad N-Channel MOSFET Drives for Two Synchronous Rectified Bridges
- Adaptive Shoot-Through Protection
 - Active Gate Threshold Monitoring
 - Programmable Dead-Time
- 0.4Ω ON-Resistance and 4A Sink Current Capability
- Supports High Switching Frequency
 - Fast Output Rise and Fall
 - Ultra Low Three-State Hold-Off Time (20ns)
- Low V_F Internal Bootstrap Diode
- Low Bias Supply Current
- Power-On Reset
- QFN Package
 - Compliant to JEDEC PUB95 MO-220 QFN-Quad Flat No Leads-Product Outline
 - Near Chip-Scale Package Footprint; Improves PCB Efficiency and Thinner in Profile
- Pb-Free Available (RoHS Compliant)

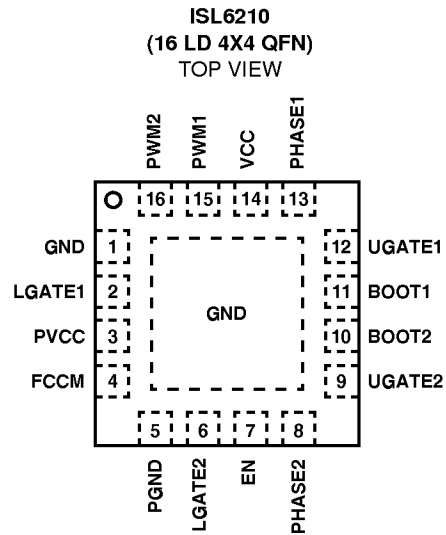
Applications

- Core Voltage Supplies for Intel® and AMD® Microprocessors
- High Frequency Low Profile High Efficiency DC/DC Converters
- High Current Low Voltage DC/DC Converters
- Synchronous Rectification for Isolated Power Supplies

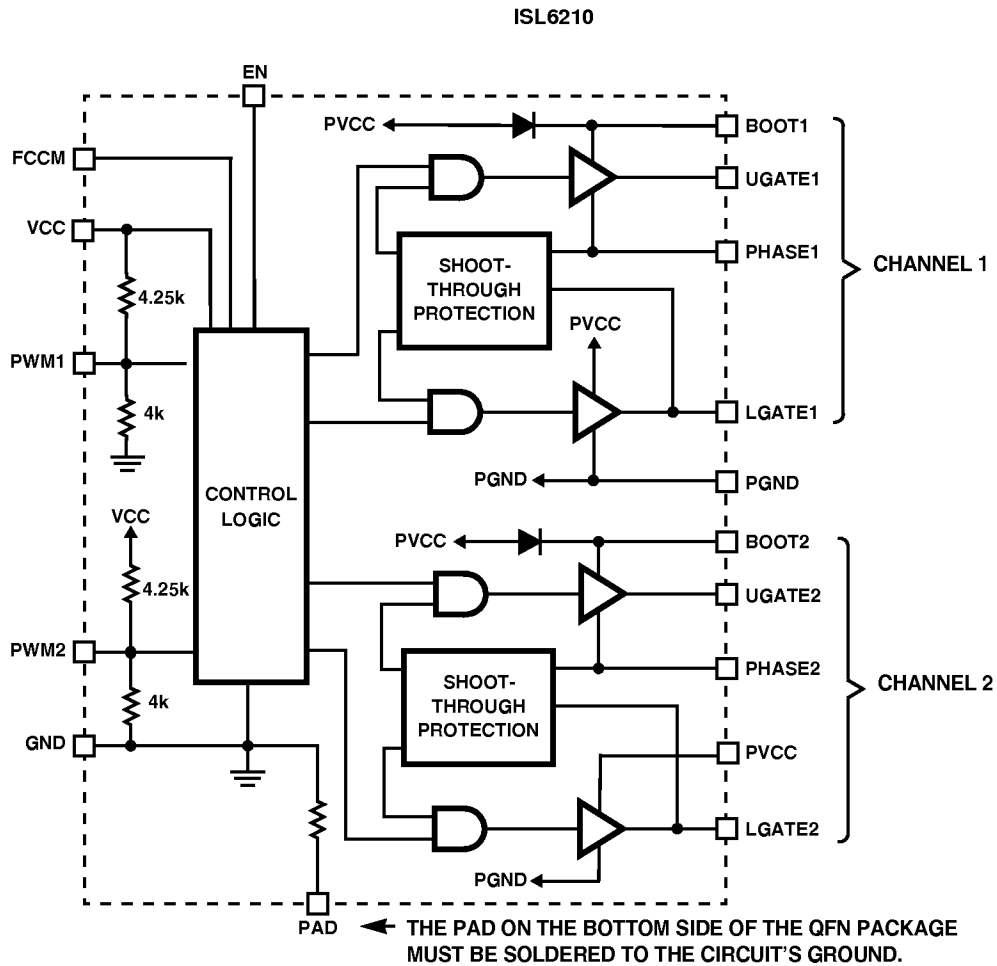
Related Literature

- Technical Brief TB363 "Guidelines for Handling and Processing Moisture Sensitive Surface Mount Devices (SMDs)"
- Technical Brief 400 and Technical Brief 417 for Power Train Design, Layout Guidelines, and Feedback Compensation Design
- Technical Brief 447 "Guidelines for Preventing Boot-to-Phase Stress on Half-Bridge MOSFET Driver ICs"

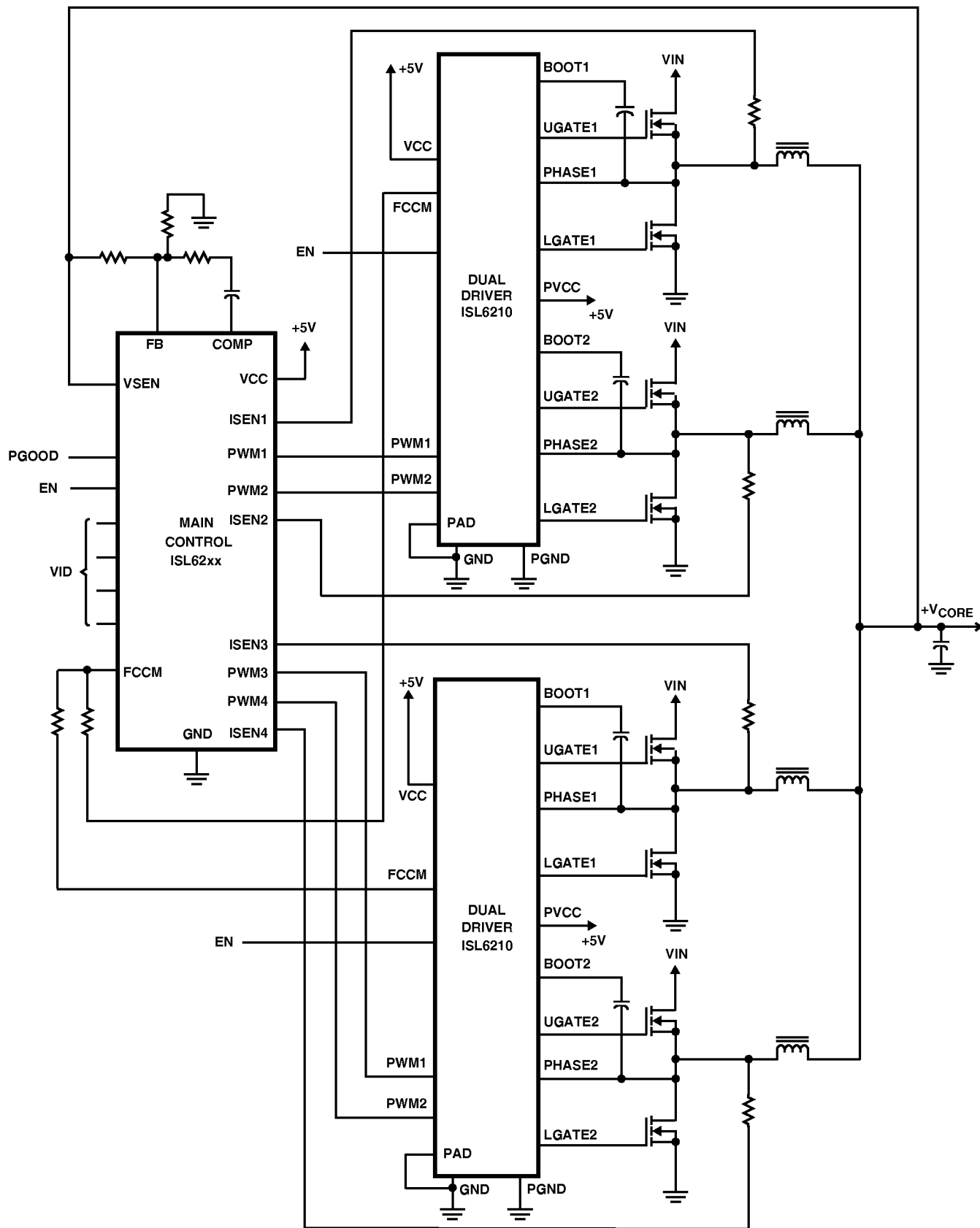
Pinout



Block Diagram



Typical Application - Multiphase Converter Using ISL6210 Gate Drivers



Absolute Maximum Ratings

Supply Voltage (PVCC, VCC)	-0.3V to 7V
Input Voltage (V _{EN} , V _{PWM})	-0.3V to VCC + 0.3V
BOOT Voltage (V _{BOOT-GND})	-0.3V to 33V (DC) or 36V (<200ns)
BOOT To PHASE Voltage (V _{BOOT-PHASE})	-0.3V to 7V (DC) -0.3V to 9V (<10ns)
PHASE Voltage	GND - 0.3V to 30V (DC) GND - 8V (<20ns Pulse Width, 10μJ)
UGATE Voltage	V _{PHASE} - 0.3V (DC) to V _{BOOT} V _{PHASE} - 5V (<20ns Pulse Width, 10μJ) to V _{BOOT}
LGATE Voltage	GND - 0.3V (DC) to VCC + 0.3V GND - 2.5V (<20ns Pulse Width, 5μJ) to VCC + 0.3V
Ambient Temperature Range	-40°C to +125°C

CAUTION: Do not operate at or near the maximum ratings listed for extended periods of time. Exposure to such conditions may adversely impact product reliability and result in failures not covered by warranty.

NOTES:

1. θ_{JA} is measured in free air with the component mounted on a high effective thermal conductivity test board with “direct attach” features.
2. θ_{JC} , “case temperature” location is at the center of the package underside exposed pad. See Tech Brief TB379 for details.

Thermal Information

Thermal Resistance (Notes 1 and 2)	θ_{JA} (°C/W)	θ_{JC} (°C/W)
QFN Package	46	8.5
Maximum Junction Temperature	+150°C	
Maximum Storage Temperature Range	-65°C to +150°C	
Pb-Free Reflow Profile	see link below http://www.intersil.com/pbfree/Pb-FreeReflow.asp	

Recommended Operating Conditions

Ambient Temperature Range	-10°C to +100°C
Maximum Operating Junction Temperature	+125°C
Supply Voltage, VCC	5V ±10%

Electrical Specifications

These specifications apply for T_A = -10°C to +100°C, Unless Otherwise Noted. Parameters with MIN and/or MAX limits are 100% tested at +25°C, unless otherwise specified. Temperature limits established by characterization and are not production tested.

PARAMETER	SYMBOL	TEST CONDITIONS	MIN	TYP	MAX	UNITS
SUPPLY CURRENT						
Bias Supply Current	I _{VCC}	PWM pin floating, V _{VCC} = 5V	-	170	-	μA
POWER-ON RESET						
POR Rising			-	3.4	4.2	V
POR Falling			2.6	3.0	-	V
Hysteresis			-	400	-	mV
BOOTSTRAP DIODE						
Forward Voltage Drop	V _F	V _{VCC} = 5V, forward bias current = 2mA	0.3	0.60	0.7	V
PWM INPUT						
Sinking Impedance	R _{PWM_SNK}		8.0	10.4	15	kΩ
Source Impedance	R _{PWM_SRC}		8.3	10.6	25	kΩ
Three-State Rising Threshold		V _{VCC} = 5V	1.08	1.3	1.5	V
Three-State Falling Threshold		V _{VCC} = 5V	3.4	3.65	3.98	V
Three-State Shutdown Holdoff Time	t _{TSSHD}	t _{PDLU} or t _{PDLL} + Gate Falling Time	-	80	-	ns
Three-state to UG/LG Rising Propagation Delay	t _{PTS}		-	20	-	ns
SWITCHING TIME (See Figure 1)						
UGATE Rise Time (Note 4)	t _{RU}	V _{VCC} = 5V, 3nF Load	-	8.0	-	ns
LGATE Rise Time (Note 4)	t _{RL}	V _{VCC} = 5V, 3nF Load	-	8.0	-	ns
UGATE Fall Time (Note 4)	t _{FU}	V _{VCC} = 5V, 3nF Load	-	8.0	-	ns
LGATE Fall Time (Note 4)	t _{FL}	V _{VCC} = 5V, 3nF Load	-	4.0	-	ns
UGATE Turn-Off Propagation Delay	t _{PDLU}	V _{VCC} = 5V, Outputs Unloaded	-	20	-	ns
LGATE Turn-Off Propagation Delay	t _{PDLL}	V _{VCC} = 5V, Outputs Unloaded	-	27	-	ns
UGATE Turn-On Propagation Delay	t _{PDHU}	V _{VCC} = 5V, Outputs Unloaded; R _{SET} = 0Ω	-	26	-	ns
LGATE Turn-On Propagation Delay	t _{PDHL}	V _{VCC} = 5V, Outputs Unloaded; R _{SET} = 0Ω	-	26	-	ns
UGATE Turn-On Propagation Delay	t _{PDHU}	V _{VCC} = 5V, Outputs Unloaded; R _{SET} = 80kΩ	-	41	-	ns
LGATE Turn-On Propagation Delay	t _{PDHL}	V _{VCC} = 5V, Outputs Unloaded; R _{SET} = 80kΩ	-	33	-	ns
Minimum LGATE On Time in DCM (Note 4)	t _{LGMIN}		-	400	-	ns

Electrical Specifications

These specifications apply for $T_A = -10^{\circ}\text{C}$ to $+100^{\circ}\text{C}$, Unless Otherwise Noted. **(Continued)** Parameters with MIN and/or MAX limits are 100% tested at $+25^{\circ}\text{C}$, unless otherwise specified. Temperature limits established by characterization and are not production tested.

PARAMETER	SYMBOL	TEST CONDITIONS	MIN	TYP	MAX	UNITS
OUTPUT						
Upper Drive Source Resistance	R_{UG_SRC}	250mA Source Current	-	1.0	2.5	Ω
Upper Drive Source Current (Note 4)	I_{UG_SCR}	$V_{UGATE-Phase} = 2.5V$	-	2.00	-	A
Upper Drive Sink Resistance	R_{UG_SNK}	250mA Sink Current	-	1.0	2.5	Ω
Upper Drive Sink Current (Note 4)	I_{UG_SNK}	$V_{UGATE-Phase} = 2.5V$		2.00	-	A
Lower Drive Source Resistance	R_{LG_SRC}	250mA Source Current	-	1.0	2.5	Ω
Lower Drive Source Current (Note 4)	I_{LG_SCR}	$V_{LGATE} = 2.5V$	-	2.00	-	A
Lower Drive Sink Resistance	R_{LG_SNK}	250mA Sink Current	-	0.4	1.0	Ω
Lower Drive Sink Current (Note 4)	I_{LG_SNK}	$V_{LGATE} = 2.5V$	-	4.00	-	A

NOTE:

- Parameters with MIN and/or MAX limits are 100% tested at $+25^{\circ}\text{C}$, unless otherwise specified. Temperature limits established by characterization and are not production tested.
- Limits established by characterization and are not production tested.

Functional Pin Description

NUMBER	NAME	FUNCTION
1	GND	Bias and reference ground. All signals are referenced to this node.
2	LGATE1	Lower gate drive output of Channel 1. Connect to gate of the low-side power N-Channel MOSFET.
3	PVCC	This pin supplies power to both the lower and higher gate drives in ISL6210. Connect to a +5V supply. Place a high quality low ESR ceramic capacitor from this pin to GND.
4	FCCM	Logic control input that will force continuous conduction mode (HIGH state) or allow discontinuous conduction mode (LOW state). Placing a series resistor in this input will allow the switching dead-time to be programmed.
5	PGND	It is the power ground return of both low gate drivers.
6	LGATE2	Lower gate drive output of Channel 2. Connect to gate of the low-side power N-Channel MOSFET.
7	EN	Logic control input that will enable (HIGH state) or disable (LOW state) the IC. Shutdown current is $<1\mu\text{A}$.
8	PHASE2	Connect this pin to the SOURCE of the upper MOSFET and the DRAIN of the lower MOSFET in Channel 2. This pin provides a return path for the upper gate drive.
9	UGATE2	Upper gate drive output of Channel 2. Connect to gate of high-side power N-Channel MOSFET.
10	BOOT2	Floating bootstrap supply pin for the upper gate drive of Channel 2. Connect the bootstrap capacitor between this pin and the PHASE2 pin. The bootstrap capacitor provides the charge to turn on the upper MOSFET. See "Internal Bootstrap Diode" on page 7 for guidance in choosing the capacitor value.
11	BOOT1	Floating bootstrap supply pin for the upper gate drive of Channel 1. Connect the bootstrap capacitor between this pin and the PHASE1 pin. The bootstrap capacitor provides the charge to turn on the upper MOSFET. See "Internal Bootstrap Diode" on page 7 for guidance in choosing the capacitor value.
12	UGATE1	Upper gate drive output of Channel 1. Connect to gate of high-side power N-Channel MOSFET.
13	PHASE1	Connect this pin to the SOURCE of the upper MOSFET and the DRAIN of the lower MOSFET in Channel 1. This pin provides a return path for the upper gate drive.
14	VCC	Connect a +5V bias supply to this pin. It supplies the internal analog circuits. Place a high quality, low ESR ceramic capacitor from this pin to GND. This should be a separate capacitor than the one used for PVCC (Pin 3).
15	PWM1	The PWM signal is the control input for the Channel 1 driver. The PWM signal can enter three distinct states during operation. See "Three-State PWM Input" on page 6 for further details. Connect this pin to the PWM output of the controller.
16	PWM2	The PWM signal is the control input for the Channel 2 driver. The PWM signal can enter three distinct states during operation. See "Three-State PWM Input" on page 6 for further details. Connect this pin to the PWM output of the controller.
N/A	PAD	Connect this pad to the power ground plane (GND) via thermally enhanced connection.

Timing Diagram

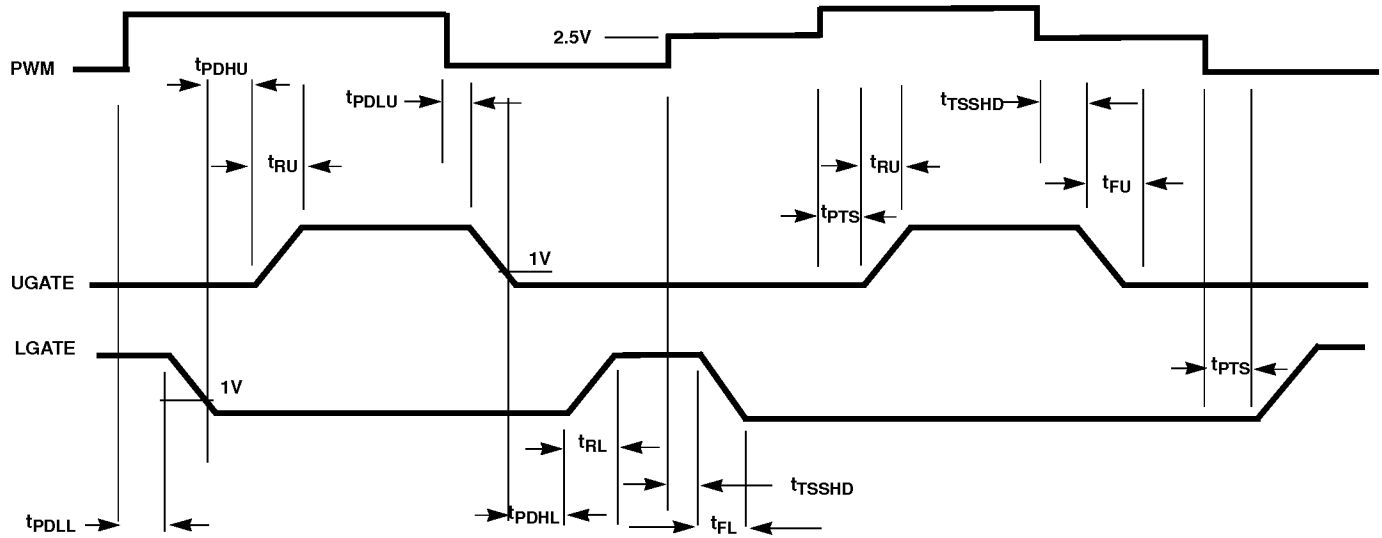


FIGURE 1. TIMING DIAGRAM

Description

Theory of Operation

Designed for speed, the ISL6210 dual MOSFET driver controls both high-side and low-side N-Channel FETs for two separate channels of a Multiphase PWM system from two independent PWM signals.

A rising edge on PWM initiates the turn-off of the lower MOSFET (see "Timing Diagram" on page 6). After a short propagation delay [t_{PDLL}], the lower gate begins to fall. Typical fall times [t_{FL}] are provided in the "Electrical Specifications" table on page 4. Adaptive shoot-through circuitry monitors the LGATE voltage. When LGATE has fallen below 1V, UGATE is allowed to turn ON. This prevents both the lower and upper MOSFETs from conducting simultaneously, or shoot-through.

A falling transition on PWM indicates the turn-off of the upper MOSFET and the turn-on of the lower MOSFET. A short propagation delay [t_{PDLU}] is encountered before the upper gate begins to fall [t_{FU}]. The upper MOSFET gate-to-source voltage is monitored, and the lower gate is allowed to rise after the upper MOSFET gate-to-source voltage drops below 1V. The lower gate then rises [t_{RL}], turning on the lower MOSFET.

This driver is optimized for converters with large step down compared to the upper MOSFET because the lower MOSFET conducts for a much longer time in a switching period. The lower gate driver is therefore sized much larger to meet this application requirement.

The 0.5Ω ON-resistance and 4A sink current capability enable the lower gate driver to absorb the current injected to the lower gate through the drain-to-gate capacitor of the

lower MOSFET and prevent a shoot through caused by the high dv/dt of the phase node.

Diode Emulation

Diode emulation allows for higher converter efficiency under light-load situations. With diode emulation active, the ISL6210 will detect the zero current crossing of the output inductor and turn off LGATE. This ensures that discontinuous conduction mode (DCM) is achieved. Diode emulation is asynchronous to the PWM signal. Therefore, the ISL6210 will respond to the FCCM input immediately after it changes state.

Please note that Intersil does not recommend Diode Emulation use with $r_{DS(ON)}$ current sensing topologies. The turn-OFF of the low side MOSFET can cause gross current measurement inaccuracies.

Three-State PWM Input

A unique feature of the ISL6210 and other Intersil drivers is the addition of a shutdown window to the PWM input. If the PWM signal enters and remains within the shutdown window for a set holdoff time, the output drivers are disabled and both MOSFET gates are pulled and held low. The shutdown state is removed when the PWM signal moves outside the shutdown window. Otherwise, the PWM rising and falling thresholds outlined in the "Electrical Specifications" table on page 4 determine when the lower and upper gates are enabled.

Adaptive Shoot-Through Protection

Both drivers incorporate adaptive shoot-through protection to prevent upper and lower MOSFETs from conducting simultaneously and shorting the input supply. This is accomplished by ensuring the falling gate has turned off one MOSFET before the other is allowed to turn on.

During turn-off of the lower MOSFET, the LGATE voltage is monitored until it reaches a 1V threshold, at which time the UGATE is released to rise. Adaptive shoot-through circuitry monitors the upper MOSFET gate-to-source voltage during UGATE turn-off. Once the upper MOSFET gate-to-source voltage has dropped below a threshold of 1V, the LGATE is allowed to rise.

In addition to gate threshold monitoring, a programmable delay between MOSFET switching can be accomplished by placing a resistor in series with the FCCM input. This delay allows for maximum design flexibility over MOSFET selection. The delay can be programmed from 5ns to 50ns and is obtained from the absolute value of the current flowing into the FCCM pin. If no resistor is used, the minimum 5ns delay is selected. Gate threshold monitoring is not affected by the addition or removal of the additional dead-time. Refer to Figure 2 and Figure 3 for more detail.

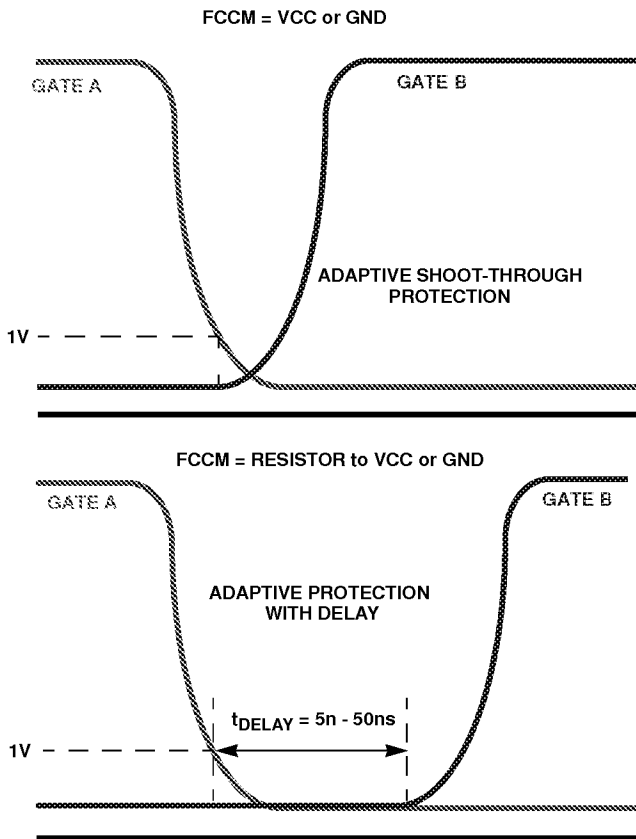


FIGURE 2. PROGRAMMABLE DEAD-TIME

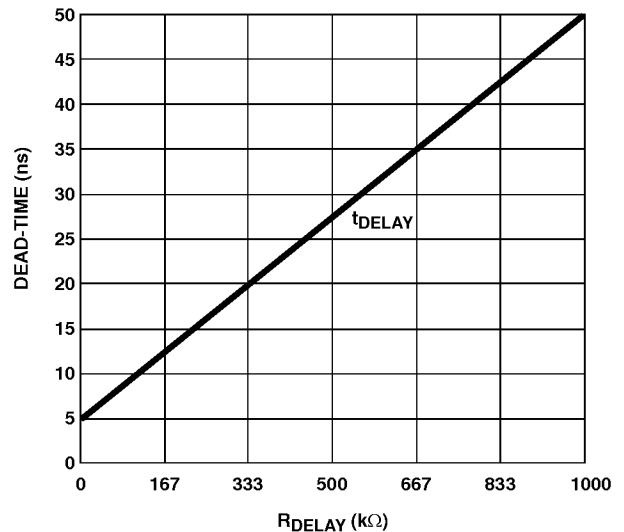


FIGURE 3. ISL6210 PROGRAMMABLE DEAD-TIME vs DELAY RESISTOR

The equation governing the dead-time seen in Figure 3 is expressed in Equation 1:

$$t_{DELAY(ns)} = [0.045 \times R_{DELAY(k\Omega)}] + 5ns \quad (EQ. 1)$$

The equation can be rewritten to solve for R_{DELAY} as follows:

$$R_{DELAY(k\Omega)} = \frac{(t_{DELAY(ns)} - 5ns)}{0.045} \quad (EQ. 2)$$

Internal Bootstrap Diode

This driver features an internal bootstrap diode. Simply adding an external capacitor across the BOOT and PHASE pins completes the bootstrap circuit.

Equation 3 helps select a proper bootstrap capacitor size:

$$C_{BOOT_CAP} \geq \frac{Q_{GATE}}{\Delta V_{BOOT_CAP}} \quad (EQ. 3)$$

$$Q_{GATE} = \frac{Q_{G1} \cdot PVCC}{V_{GS1}} \cdot N_{Q1}$$

where Q_{G1} is the amount of gate charge per upper MOSFET at V_{GS1} gate-source voltage and N_{Q1} is the number of control MOSFETs. The ΔV_{BOOT_CAP} term is defined as the allowable droop in the rail of the upper gate drive.

As an example, suppose two IRLR7821 FETs are chosen as the upper MOSFETs. The gate charge, Q_G , from the data sheet is 10nC at 4.5V (V_{GS}) gate-source voltage. Then the Q_{GATE} is calculated to be 22nC at PVCC level. We will assume a 200mV droop in drive voltage over the PWM cycle. We find that a bootstrap capacitance of at least 0.110μF is required. The next larger standard value capacitance is 0.22μF. A good quality ceramic capacitor is recommended.

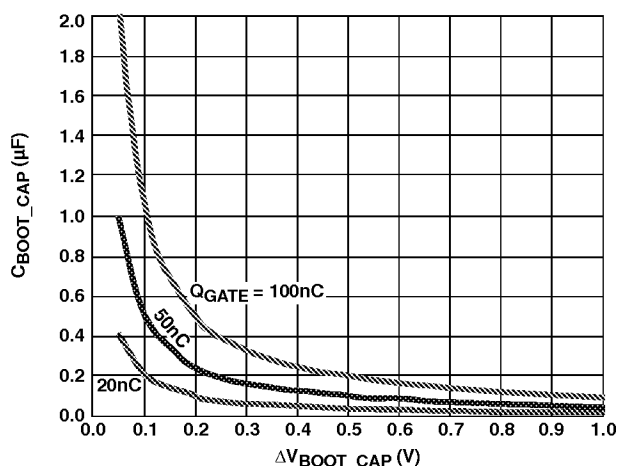


FIGURE 4. BOOTSTRAP CAPACITANCE vs BOOT RIPPLE VOLTAGE

Power Dissipation

Package power dissipation is mainly a function of the switching frequency (F_{SW}), the output drive impedance, the external gate resistance, and the selected MOSFET's internal gate resistance and total gate charge. Calculating the power dissipation in the driver for a desired application is critical to ensure safe operation. Exceeding the maximum allowable power dissipation level will push the IC beyond the maximum recommended operating junction temperature of +125°C. The maximum allowable IC power dissipation for the SO14 package is approximately 1W at room temperature, while the power dissipation capacity in the QFN packages, with an exposed heat escape pad, is around 2W. See "Layout Considerations" on page 9 for thermal transfer improvement suggestions. When designing the driver into an application, it is recommended that the following calculation is used to ensure safe operation at the desired frequency for the selected MOSFETs. The total gate drive power losses due to the gate charge of MOSFETs and the driver's internal circuitry and their corresponding average driver current can be estimated with Equations 4 and 5, respectively,

$$P_{Qg_TOT} = P_{Qg_Q1} + P_{Qg_Q2} + I_Q \cdot V_{CC} \quad (\text{EQ. 4})$$

$$P_{Qg_Q1} = \frac{Q_{G1} \cdot PV_{CC}^2}{V_{GS1}} \cdot F_{SW} \cdot N_{Q1}$$

$$P_{Qg_Q2} = \frac{Q_{G2} \cdot PV_{CC}^2}{V_{GS2}} \cdot F_{SW} \cdot N_{Q2}$$

$$I_{DR} = \left(\frac{Q_{G1} \cdot N_{Q1}}{V_{GS1}} + \frac{Q_{G2} \cdot N_{Q2}}{V_{GS2}} \right) \cdot F_{SW} + I_Q \quad (\text{EQ. 5})$$

where the gate charge (Q_{G1} and Q_{G2}) is defined at a particular gate to source voltage (V_{GS1} and V_{GS2}) in the corresponding MOSFET data sheet; I_Q is the driver's total quiescent current with no load at both drive outputs; N_{Q1} and N_{Q2} are number of upper and lower MOSFETs, respectively. The $I_Q V_{CC}$ product is the quiescent power of the driver without capacitive load and is typically negligible.

The total gate drive power losses are dissipated among the resistive components along the transition path. The drive resistance dissipates a portion of the total gate drive power losses, the rest will be dissipated by the external gate resistors (R_{G1} and R_{G2} , should be a short to avoid interfering with the operation shoot-through protection circuitry) and the internal gate resistors (R_{G11} and R_{G12}) of MOSFETs. Figures 5 and 6 show the typical upper and lower gate drives turn-on transition path. The power dissipation on the driver can be roughly estimated as follows:

$$P_{DR} = P_{DR_UP} + P_{DR_LOW} + I_Q \cdot V_{CC} \quad (\text{EQ. 6})$$

$$P_{DR_UP} = \left(\frac{R_{HI1}}{R_{HI1} + R_{EXT1}} + \frac{R_{LO1}}{R_{LO1} + R_{EXT1}} \right) \cdot \frac{P_{Qg_Q1}}{2}$$

$$P_{DR_LOW} = \left(\frac{R_{HI2}}{R_{HI2} + R_{EXT2}} + \frac{R_{LO2}}{R_{LO2} + R_{EXT2}} \right) \cdot \frac{P_{Qg_Q2}}{2}$$

$$R_{EXT2} = R_{G1} + \frac{R_{G11}}{N_{Q1}} \quad R_{EXT2} = R_{G2} + \frac{R_{G12}}{N_{Q2}}$$

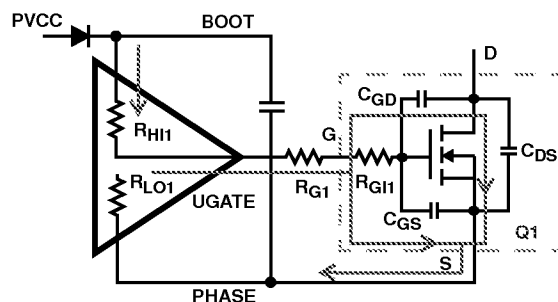


FIGURE 5. TYPICAL UPPER-GATE DRIVE TURN-ON PATH

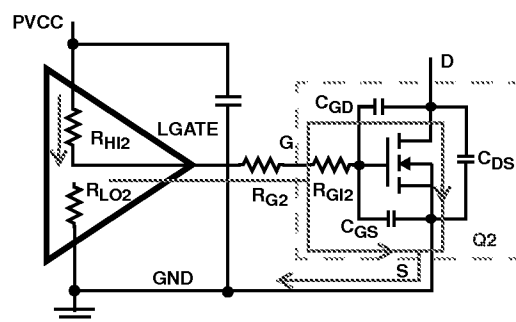


FIGURE 6. TYPICAL LOWER-GATE DRIVE TURN-ON PATH

Layout Considerations

Reducing Phase Ring

The parasitic inductances of the PCB and the power devices (both upper and lower FETs) could cause serious ringing, exceeding absolute maximum rating of the devices. The negative ringing at the edges of the PHASE node could add charges to the bootstrap capacitor through the internal bootstrap diode, in some cases, it could cause over stress across BOOT and PHASE pins. Therefore, user should do a careful layout and select proper MOSFETs and drivers. The D²PAK and DPAK package MOSFETs have high parasitic lead inductance, which can exacerbate this issue. FET selection plays an important role in reducing PHASE ring. If higher inductance FETs must be used, a Schottky diode is recommended across the lower MOSFET to clamp negative PHASE ring.

A good layout would help reduce the ringing on the phase and gate nodes significantly:

- Avoid uses via for decoupling components across BOOT and PHASE pins and in between VCC and GND pins. The decoupling loop should be short.
- All power traces (UGATE, PHASE, LGATE, GND, VCC) should be short and wide, and avoid using via; otherwise, use two vias for interconnection when possible.
- Keep SOURCE of upper FET and DRAIN of lower FET as close as thermally possible.
- Keep connection in between SOURCE of lower FET and power ground wide and short.
- Input capacitors should be placed as close to the DRAIN of upper FET and SOURCE of lower FETs as thermally possible.

NOTE: Refer to Intersil Tech Brief TB447 for more information.

Thermal Management

For maximum thermal performance in high current, high switching frequency applications, connecting the thermal pad of the QFN part to the power ground with multiple vias is recommended. This heat spreading allows the part to achieve its full thermal potential.

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Intersil Corporation's quality certifications can be viewed at www.intersil.com/design/quality

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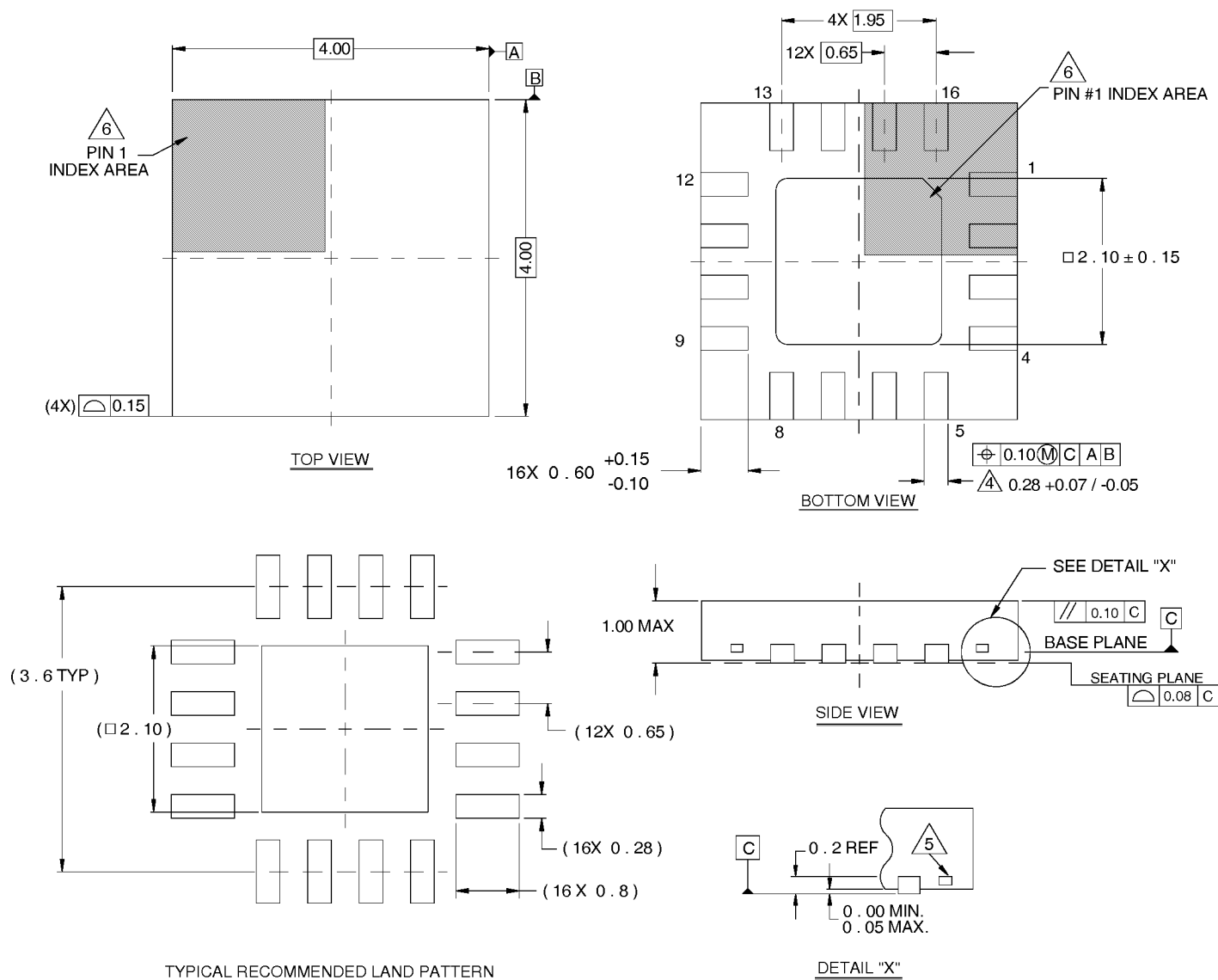
For information regarding Intersil Corporation and its products, see www.intersil.com

Package Outline Drawing

L16.4x4

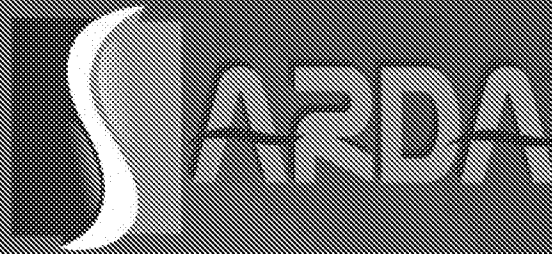
16 LEAD QUAD FLAT NO-LEAD PLASTIC PACKAGE

Rev 6, 02/08



NOTES:

- Dimensions are in millimeters.
Dimensions in () for Reference Only.
- Dimensioning and tolerancing conform to AMSE Y14.5m-1994.
- Unless otherwise specified, tolerance : Decimal ± 0.05
- Dimension b applies to the metallized terminal and is measured between 0.15mm and 0.30mm from the terminal tip.
- Tiebar shown (if present) is a non-functional feature.
- The configuration of the pin #1 identifier is optional, but must be located within the zone indicated. The pin #1 identifier may be either a mold or mark feature.



12V PowerStage in Embedded Die System-in-Package

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Agenda

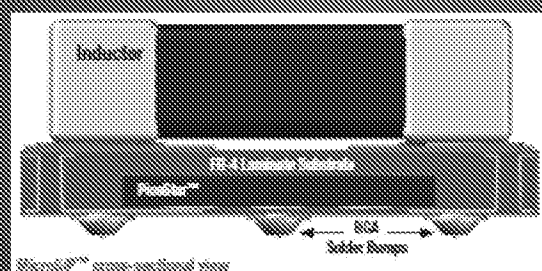
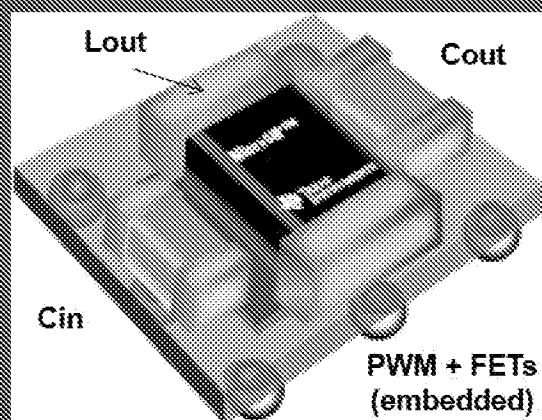
- Introduction: 3D Power Packaging
- State-of-the-Art 12V input POL Solutions
- What if you need a higher power density solution?
 - GaAs – an enabling high frequency FET
 - Multiphase – to shrink size of inductor
 - Integration – a requirement for multi-MHz switching
- Embedded Die System-in-Package approach
- Summary

3D Power Packaging is a Game Changer

using Embedded Die in Substrate Technology

In Production today

One Example: TI's MicroSIP™ Modules



Nice compact solution

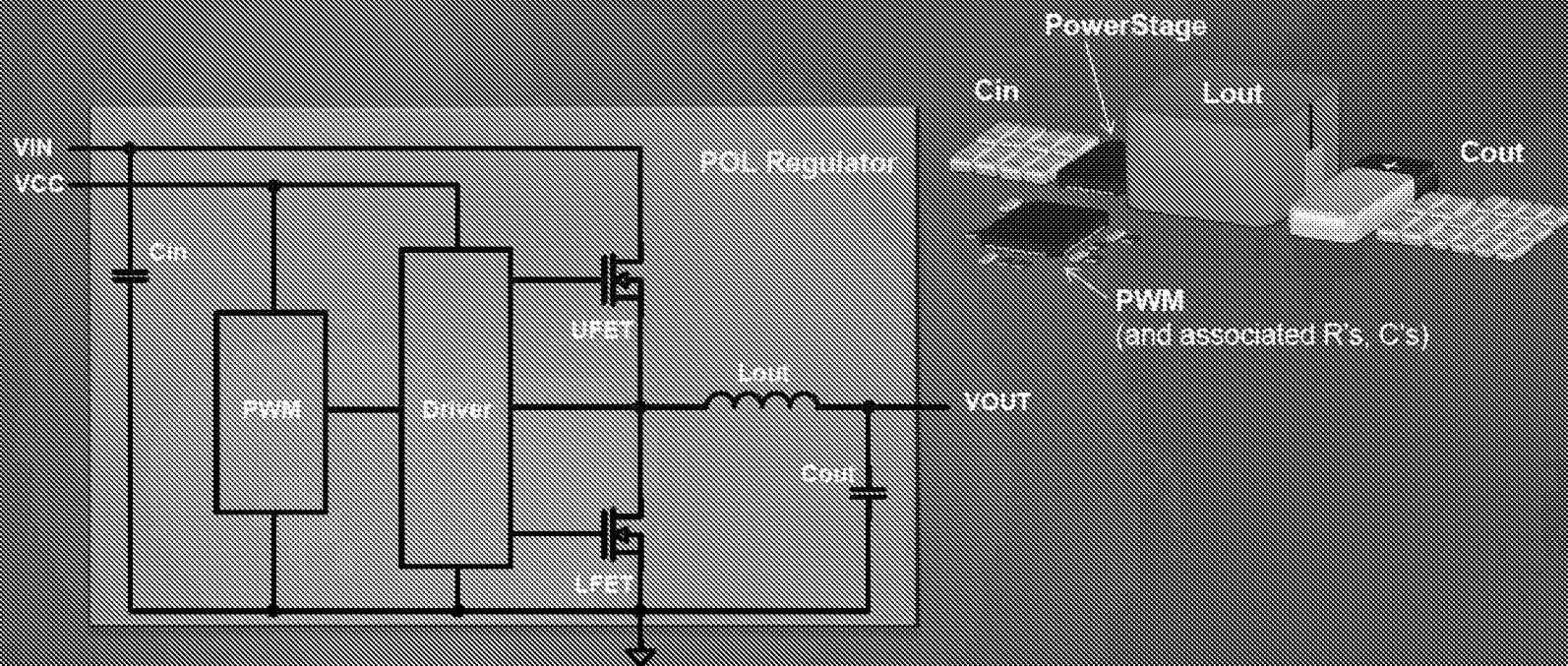
Low Input Voltage: 5V max

Low Output Current: ~2A max

*But what about higher voltages
and higher currents?*

*Does Embedded Die in Substrate
Technology add value there?*

12V Input Buck POL Regulator

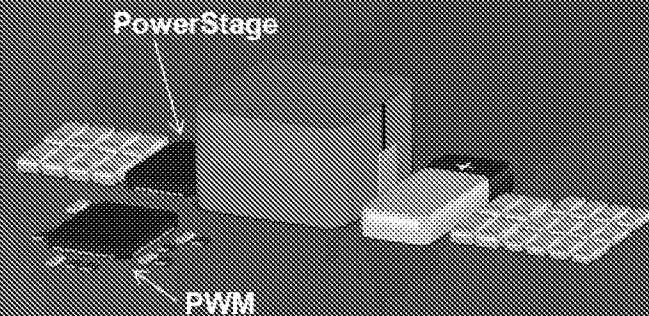
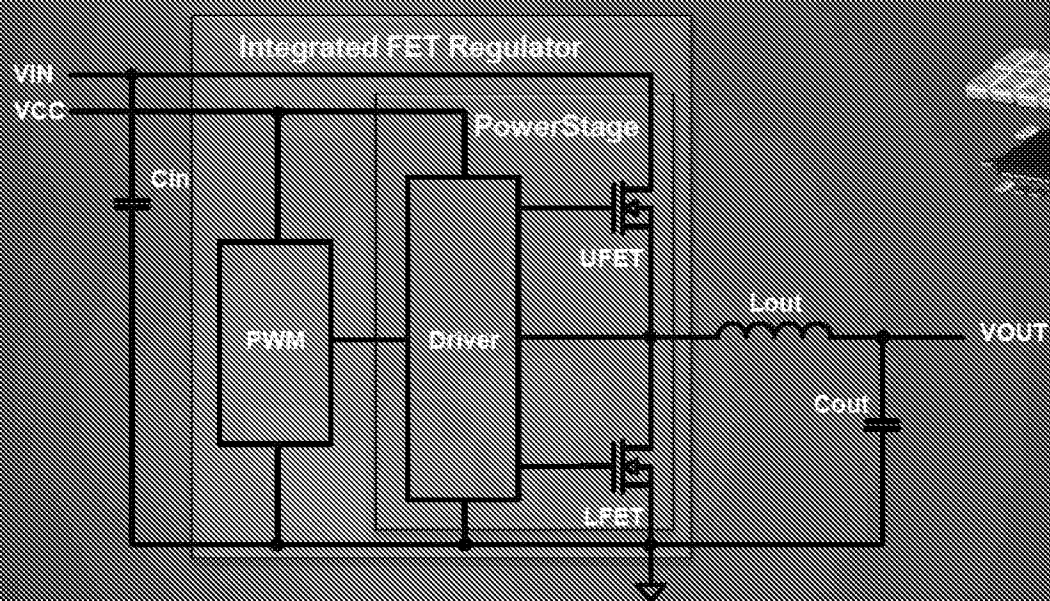


Can Embedded Die in Substrate Technology add value in 12V input, high Output Current applications?

*Answer: An emphatic Yes **

** In conjunction with Compound Semiconductor FETs, namely GaAs*

Common Integration Approaches



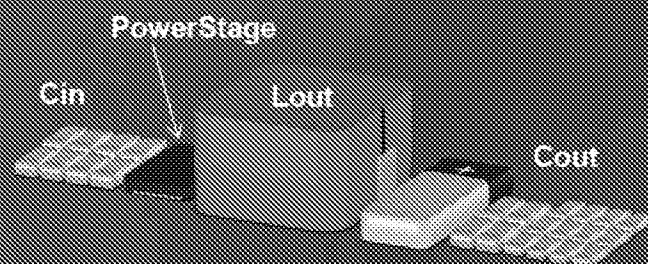
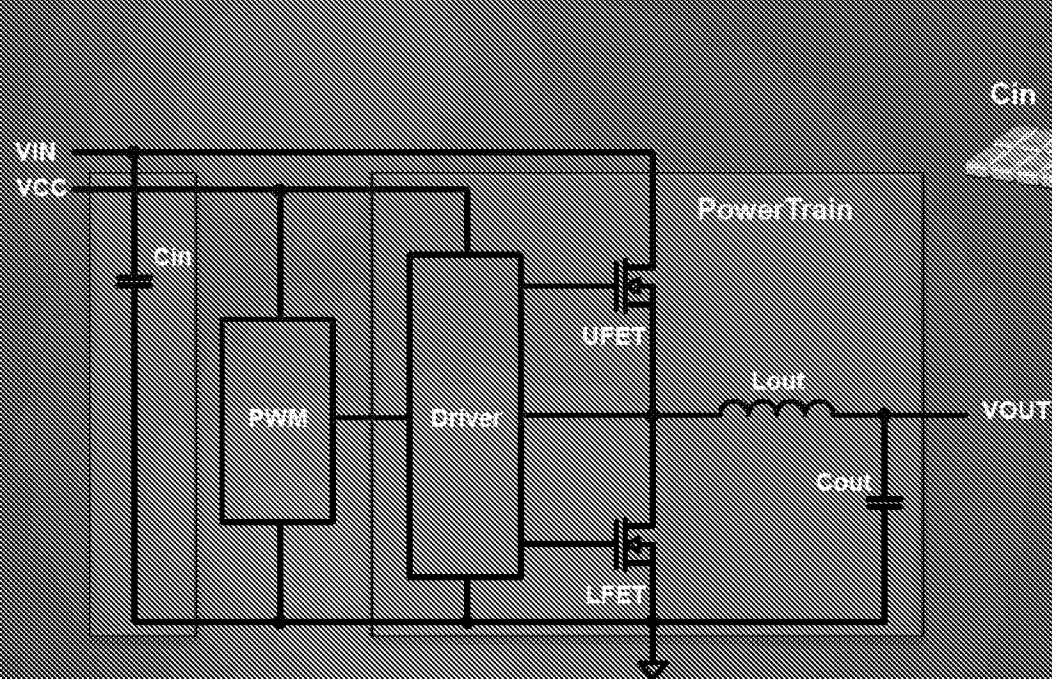
PowerStage

- Driver + FETs
- Typically multi-chip for high current applications ($>10A$)

Integrated FET Regulator

- PWM + Driver + FETs
- Typically monolithic for low current applications ($<10A$)
- Very common for low V_{in} ($\leq 5V$)

For Benchmarking Purposes – Examine PowerTrain

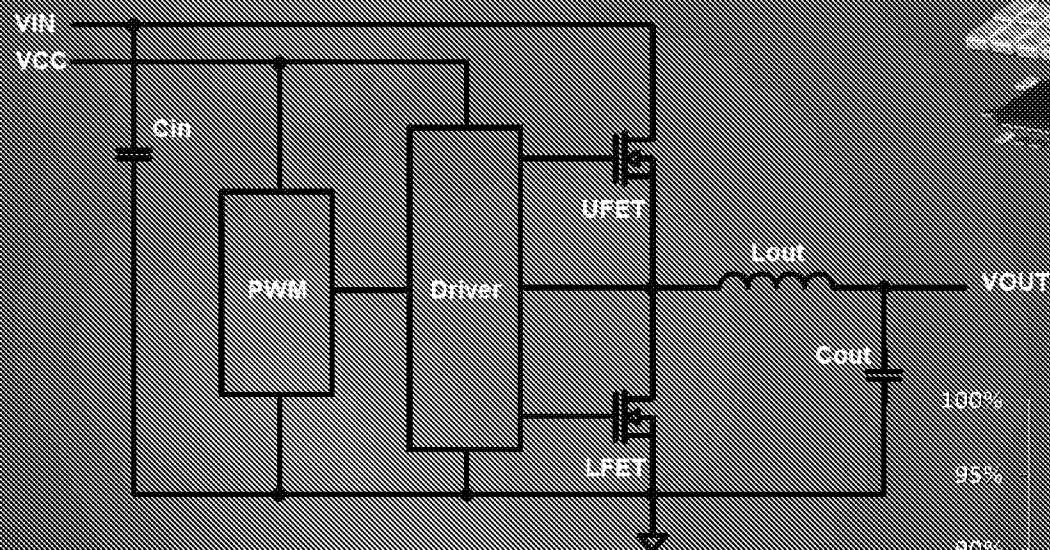


PowerTrain

- PowerStage + Output Filter + Input Cap
- Determines efficiency and majority of POL size

State-of-Art 12V input Buck POL Solution

30A output current example



Low Rdson MOSFETs

- 1mΩ LFET

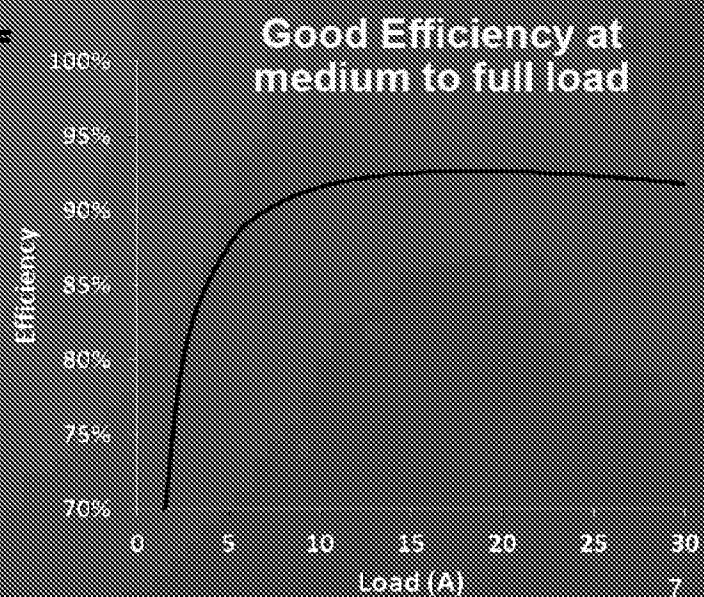
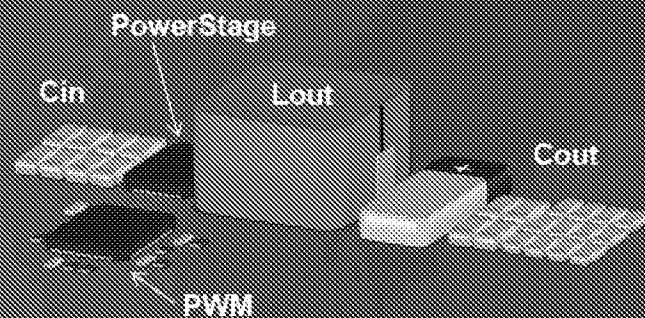
Low DCR Inductor

- 0.2mΩ

Low Switching Frequency

- 500-700kHz

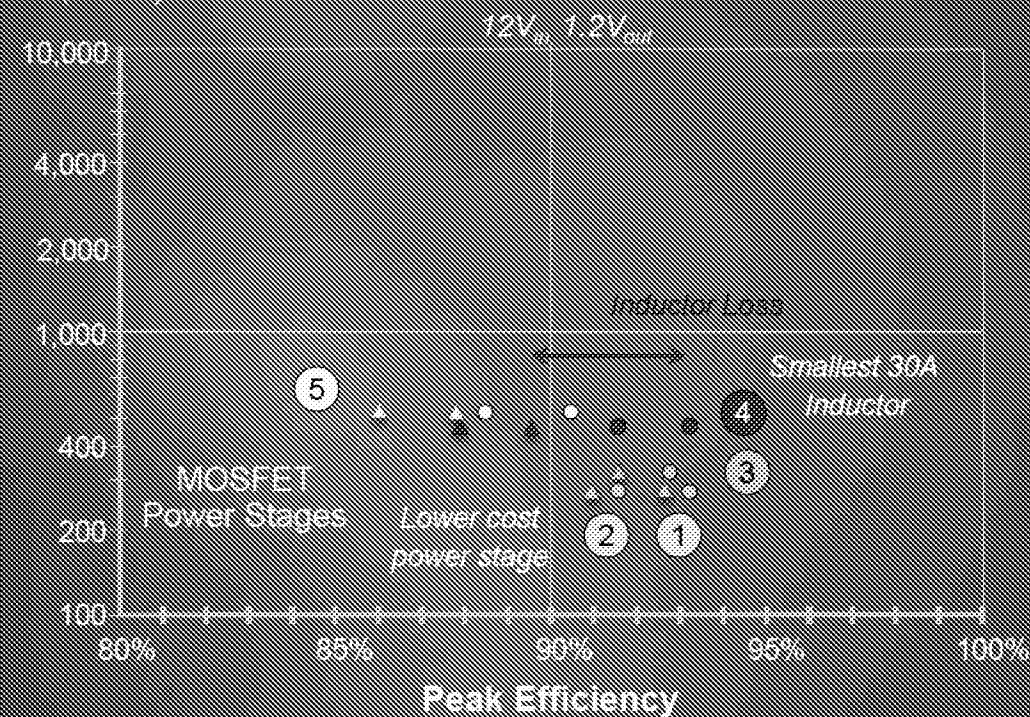
Bulky – but if you have the room for these solutions, by all means use them



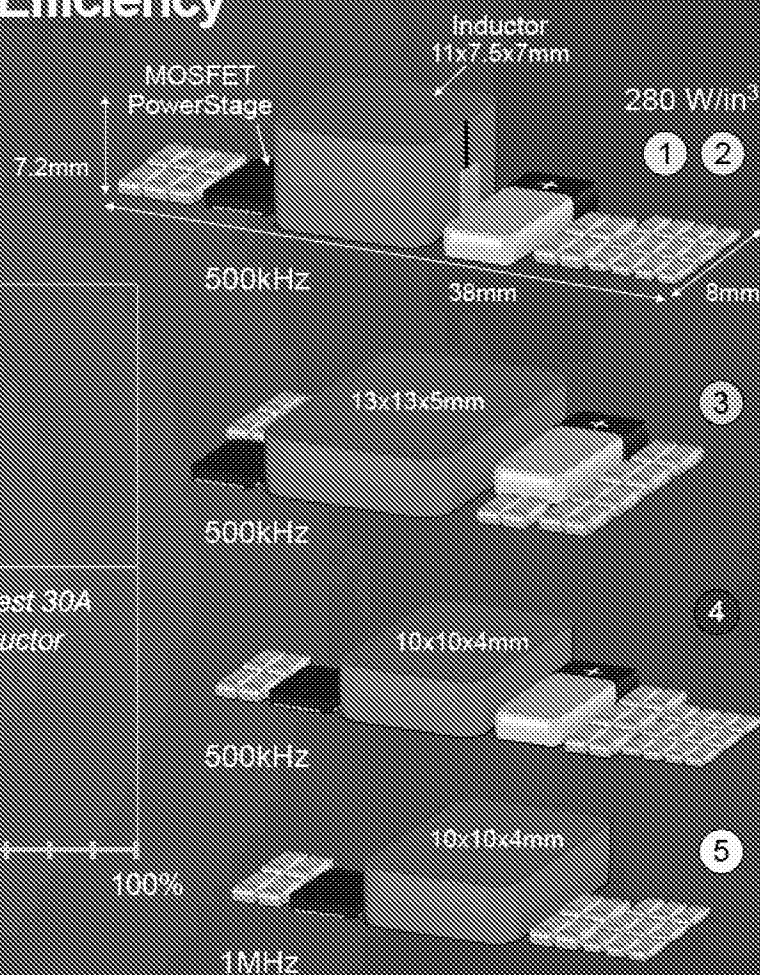
Key Metrics – Density (Size) and Efficiency

30A output current

Power Train
Density
(W/in³)



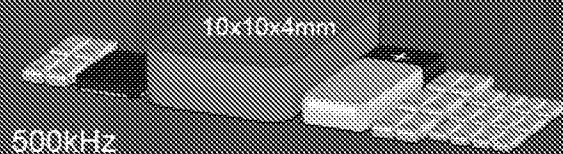
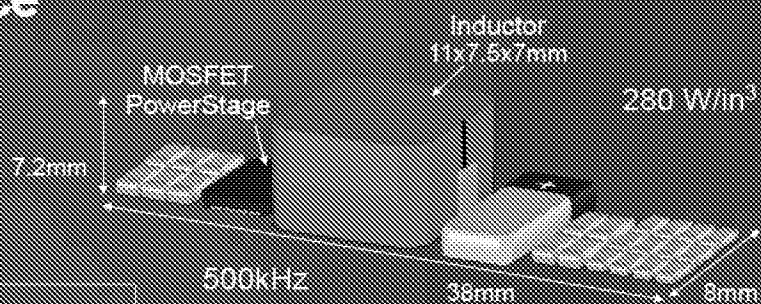
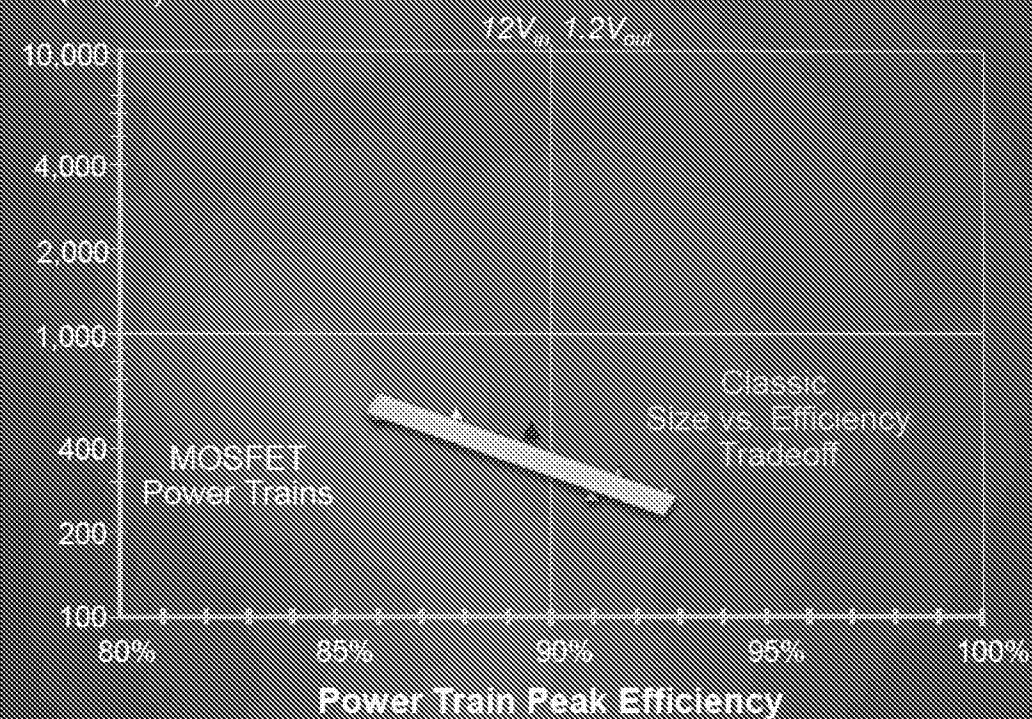
- Power Stage efficiency
- ▲ Power Train efficiency



MOSFET PowerTrain Performance

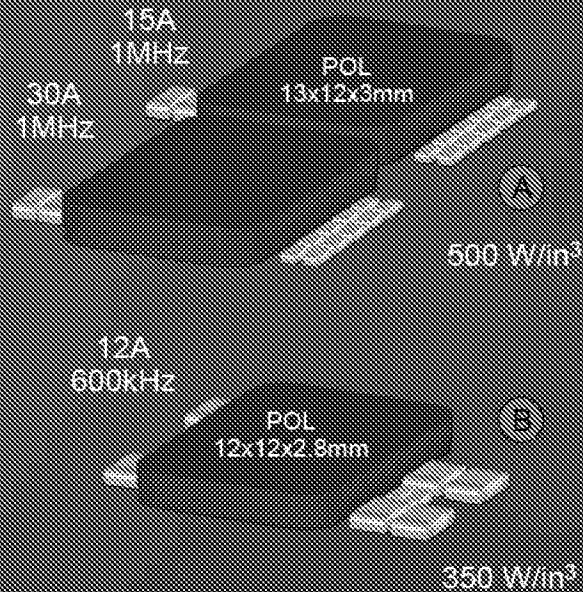
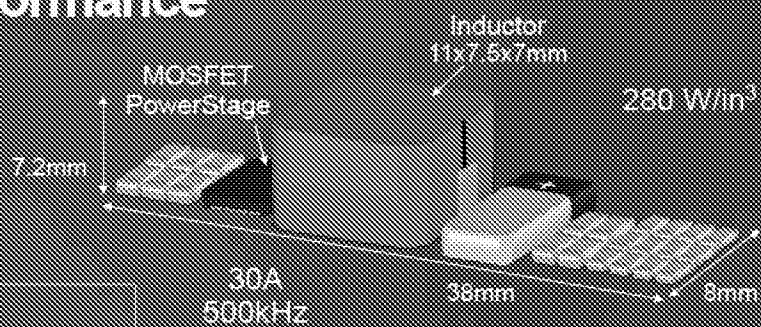
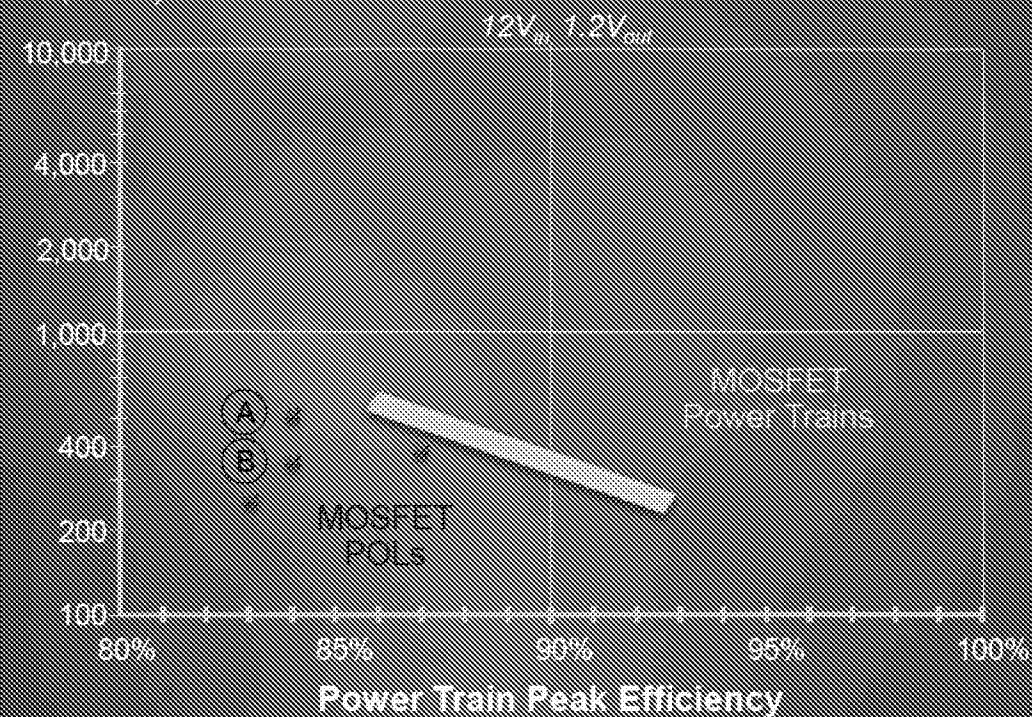
30A output current

Power Train
Density
(W/in³)



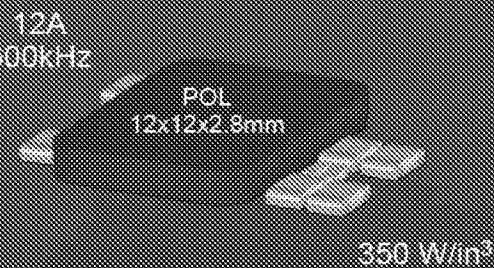
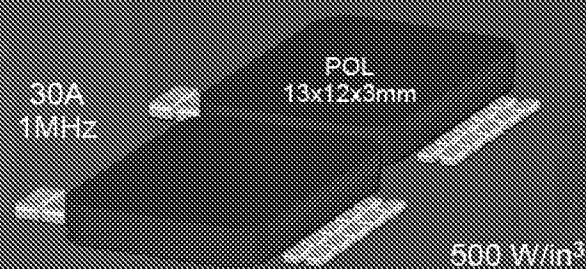
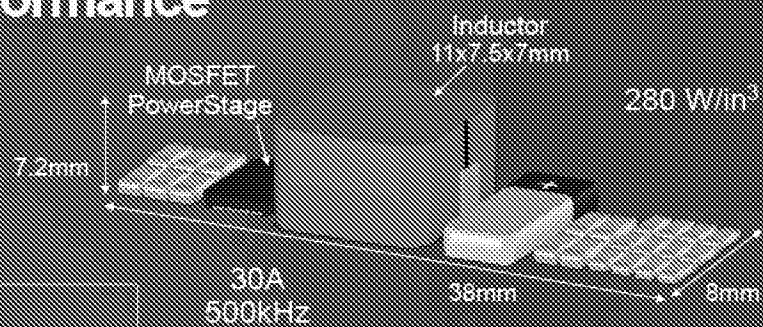
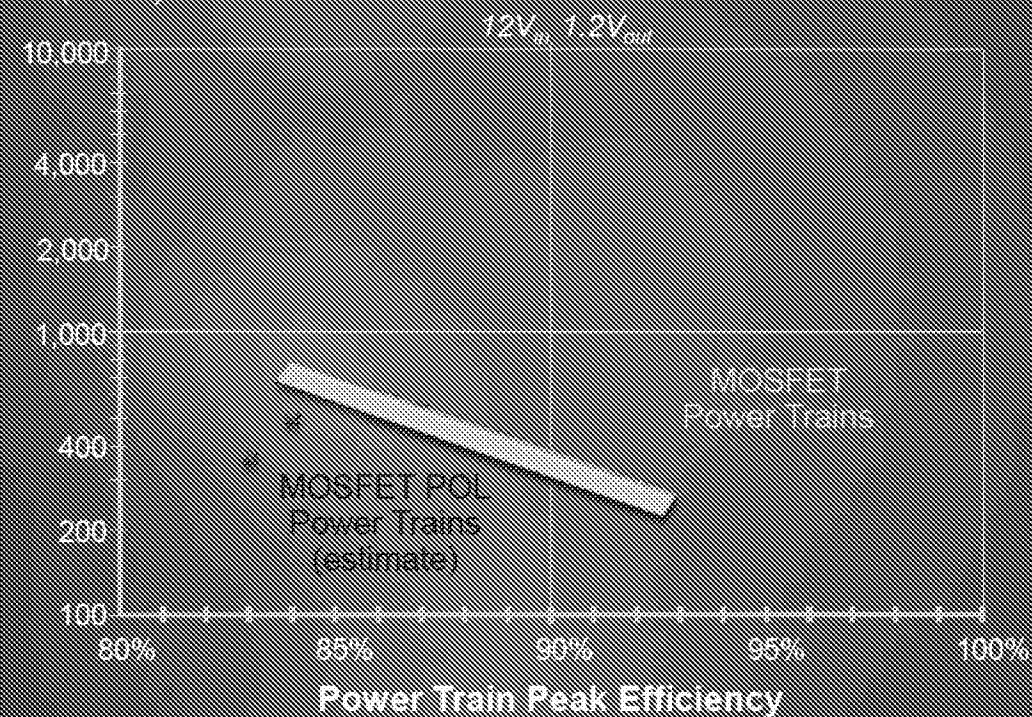
High Density POL Regulator Performance

Power Train
Density
(W/in³)



High Density POL Regulator Performance

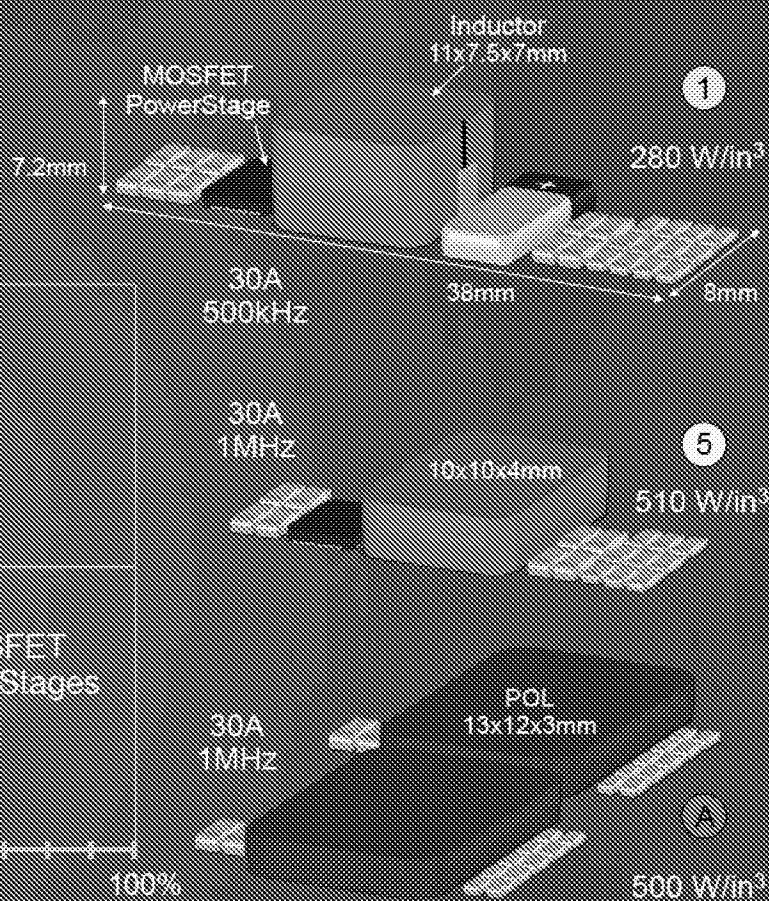
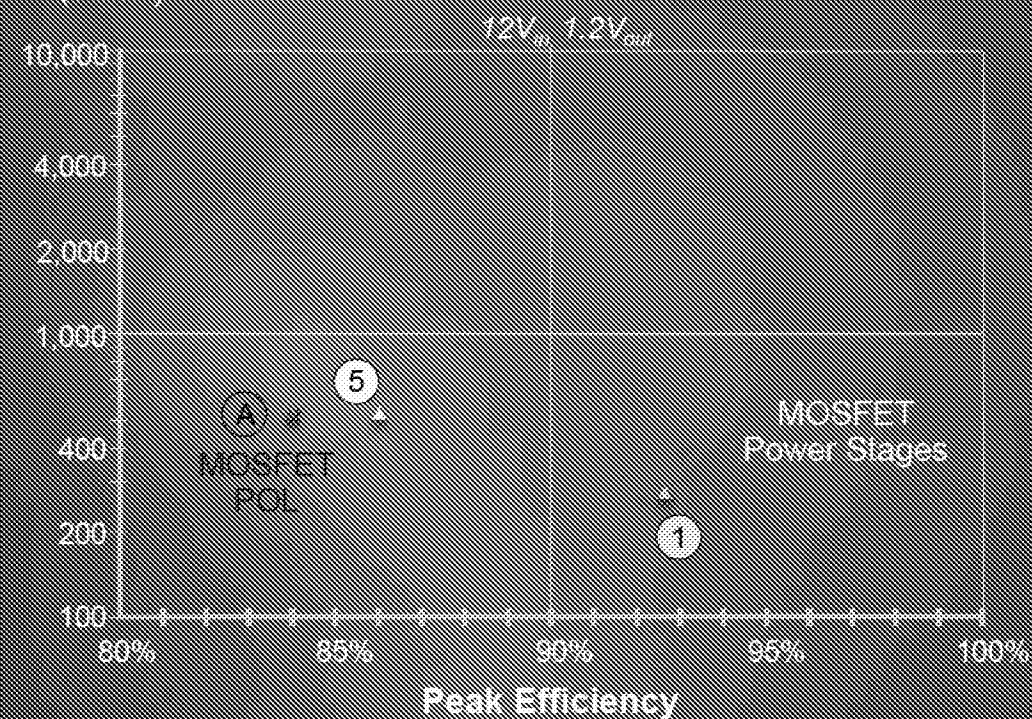
Power Train
Density
(W/in³)



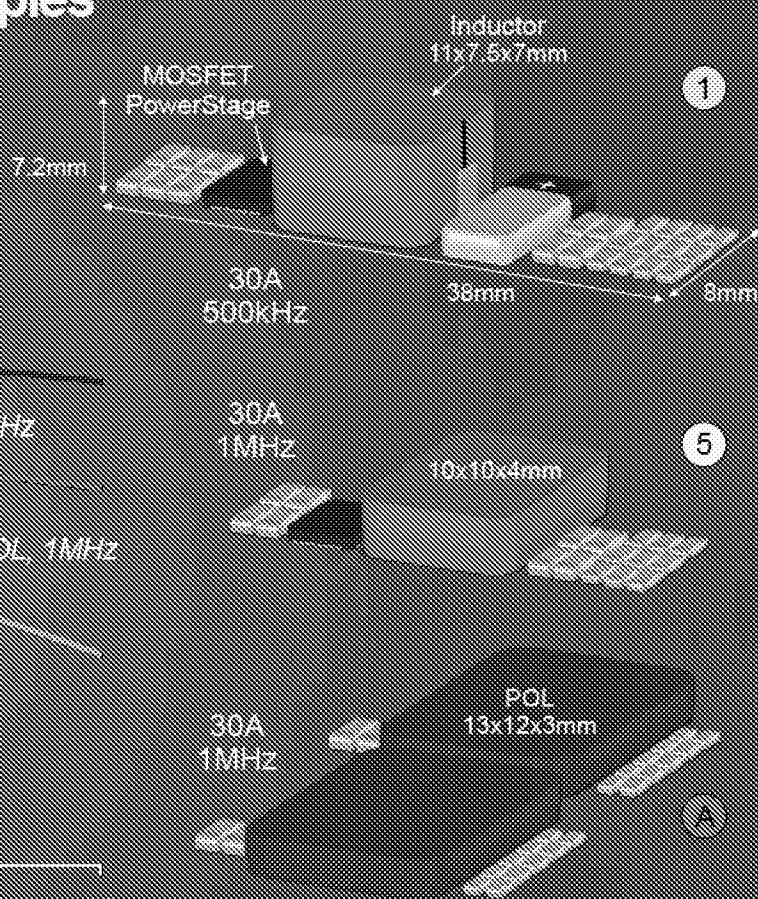
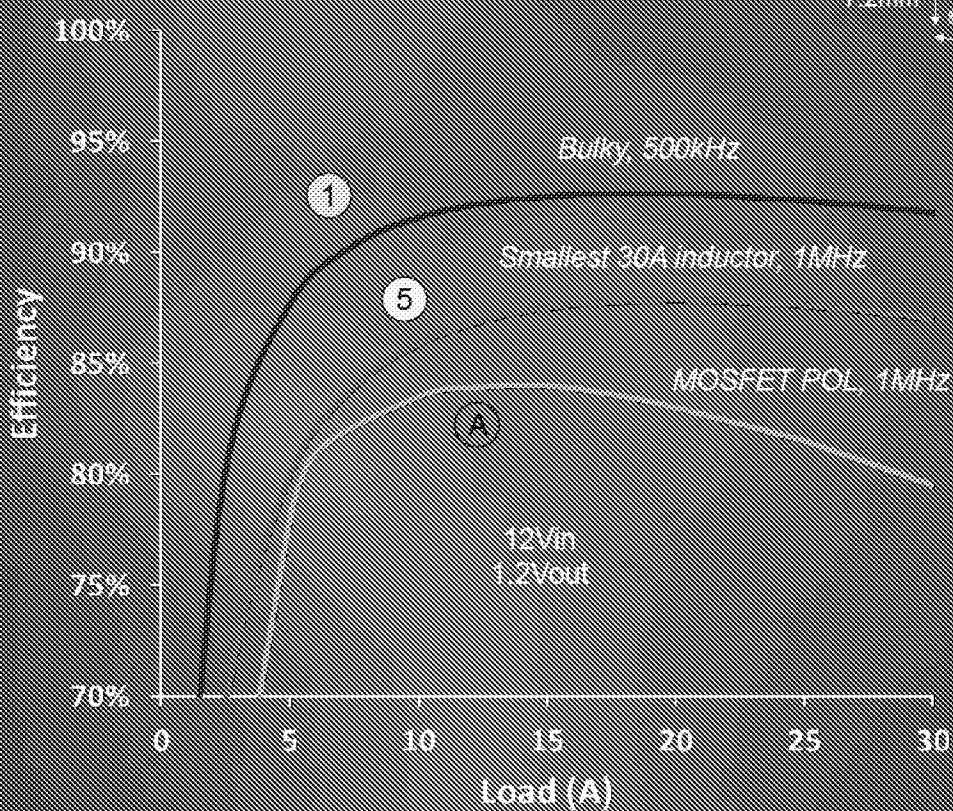
Closer Look at Three Examples

30A output current

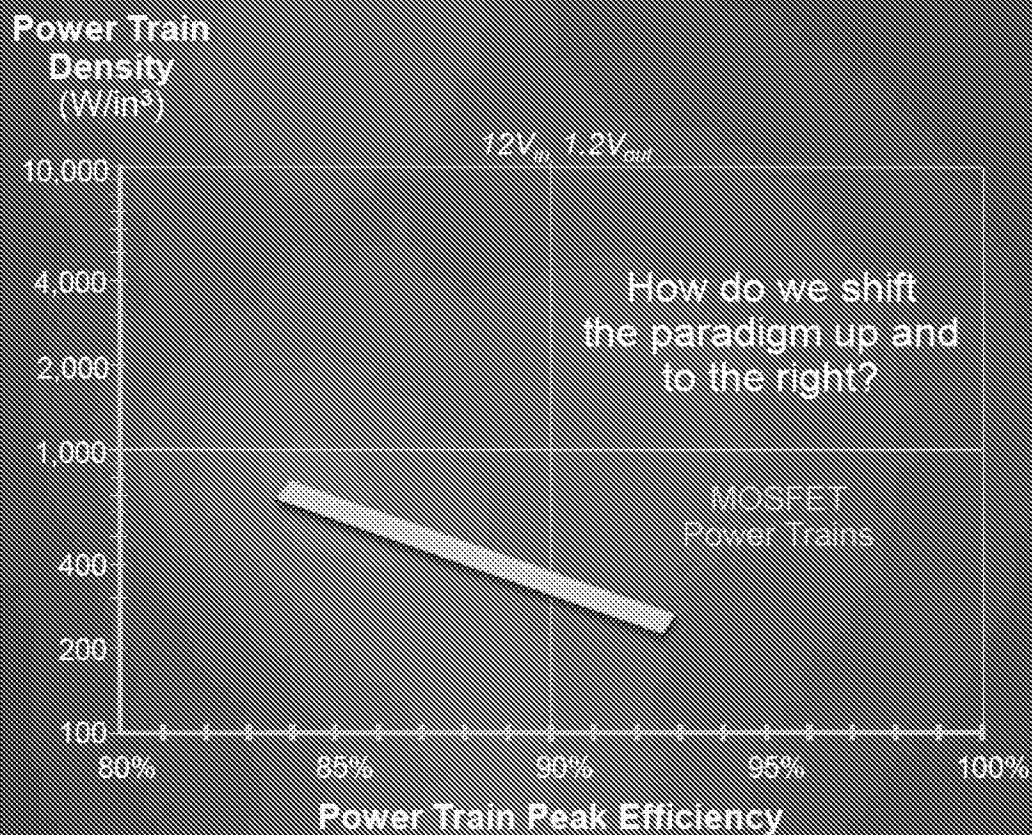
Power Train
Density
(W/in³)



Efficiency Curves of Three Examples

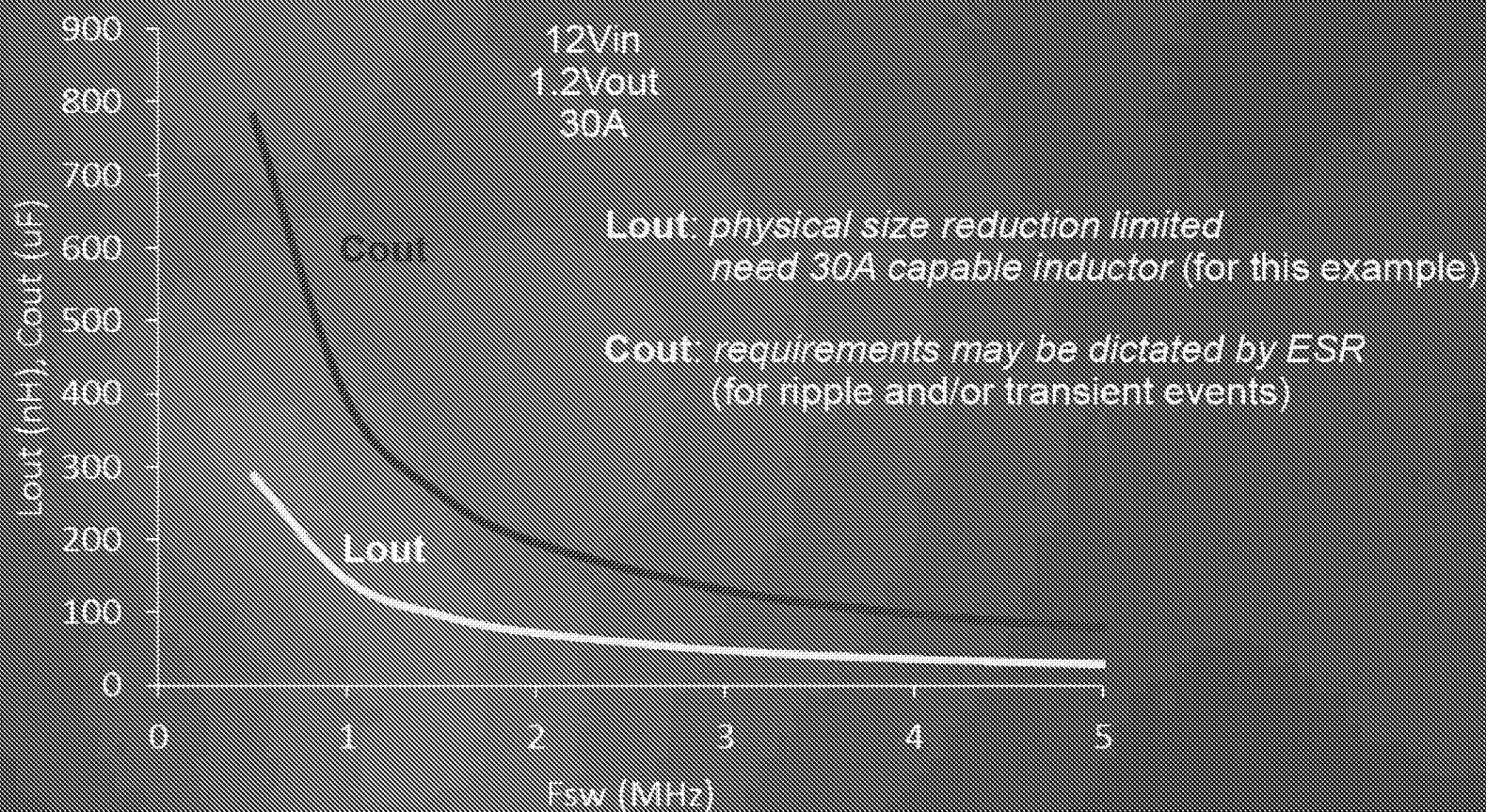


High Density POL Regulator Performance



We need higher Switching Frequency

to shrink the Output Filter

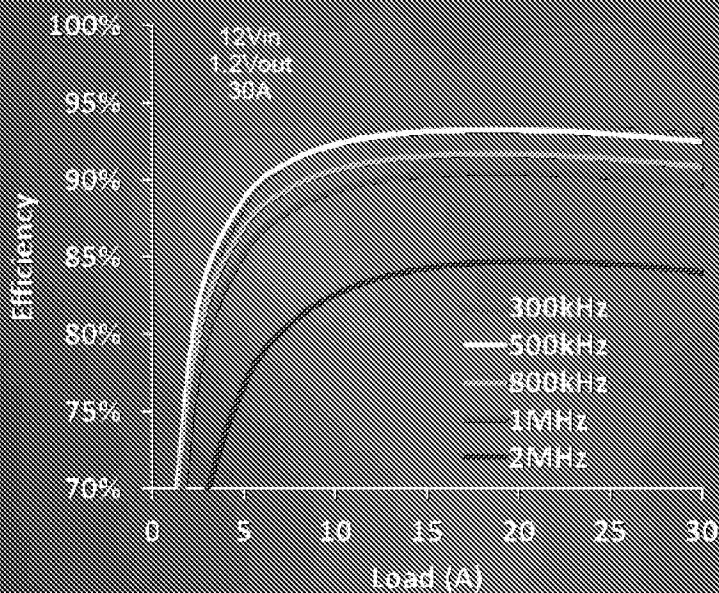


Challenges to Higher Density 12Vin POL solutions

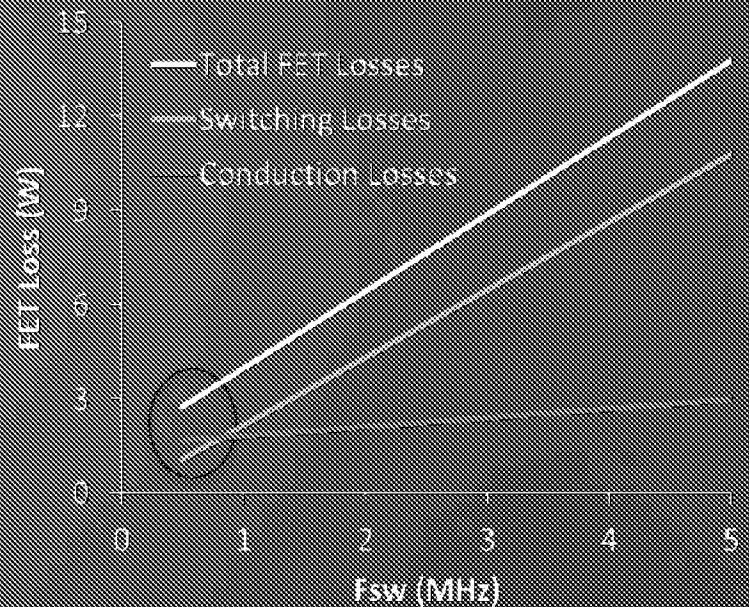
(1) FET Switching Losses

MOSFETs are not up to the challenge of multi-MHz operation

State of the art Powerstage Efficiency



State of the art Powerstage FET Losses



Limited to less than ~1MHz

GaAs FET Advantages for 12V Input POLs

	Vertical MOSFETs	GaN-on-Silicon FETs	GaAs FETs	Benefit
Electron Mobility (cm ² /Vs)	1,400	1,800	8,500	Fast Switching
R _{DS(on)} *Q _G (mΩ-nC)	~30	~25	Gen1 12 → Gen2 6	Reduced Switching loss (ability to go to higher F _{sw})
Body Diode	Yes	No	No	No Qrr loss
R _{DS(on)} temperature coefficient (% / °C)	0.4	0.2	0.2	Reduced Conduction loss
FET Structure	Vertical	Lateral	Lateral	Monolithic integration of UFET / LFET (reduced parasitics)

Green text: merit

Additional Information on Sarda's GaAs FET developments:

- * APEC 2013: "The **gFET™** Switch: A New Low Voltage High Speed GaAs HEMT for Switching Applications", Robert White, et al.
- * IMAPS 2013: "A Highly-Integrated GaAs-based Module for DC-DC Regulators", Greg Miller, et al.
- * APEC 2014: "Recent Developments In GaAs Power Switching Devices Including Device Modeling", Robert White, et al.

Challenges to Higher Density 12Vin POL solutions

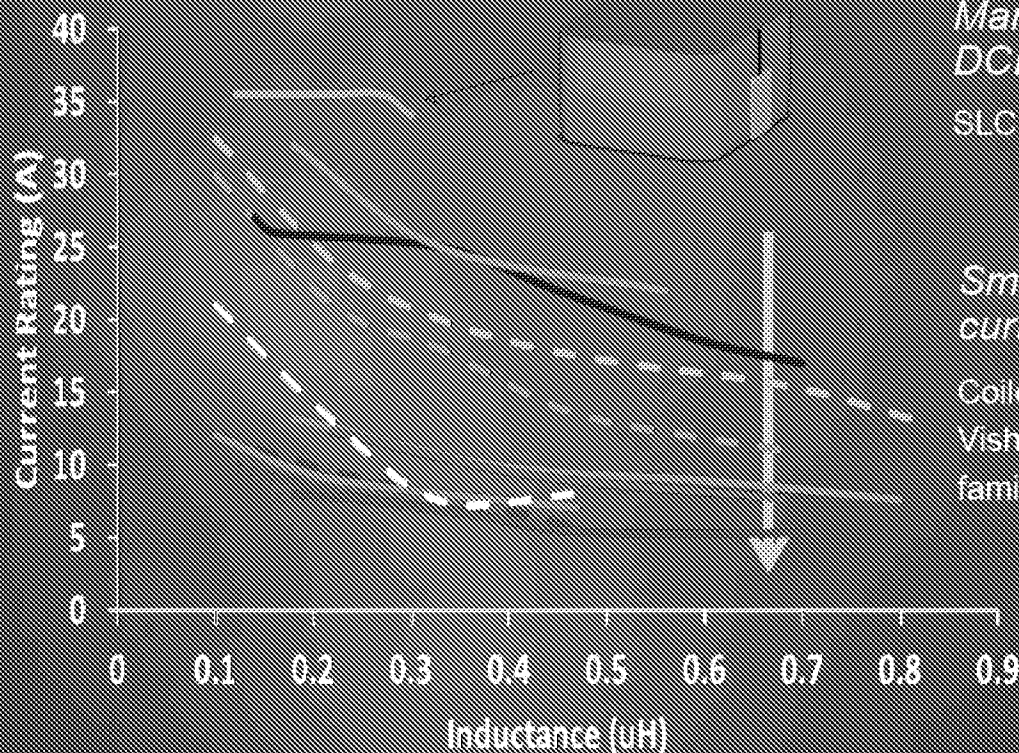
(1) FET Switching Losses → Sarda GaAs FET

(2) High current inductors are large

30A inductors are large

Many choices for low DCR, bulky Inductors

SLC1175 example



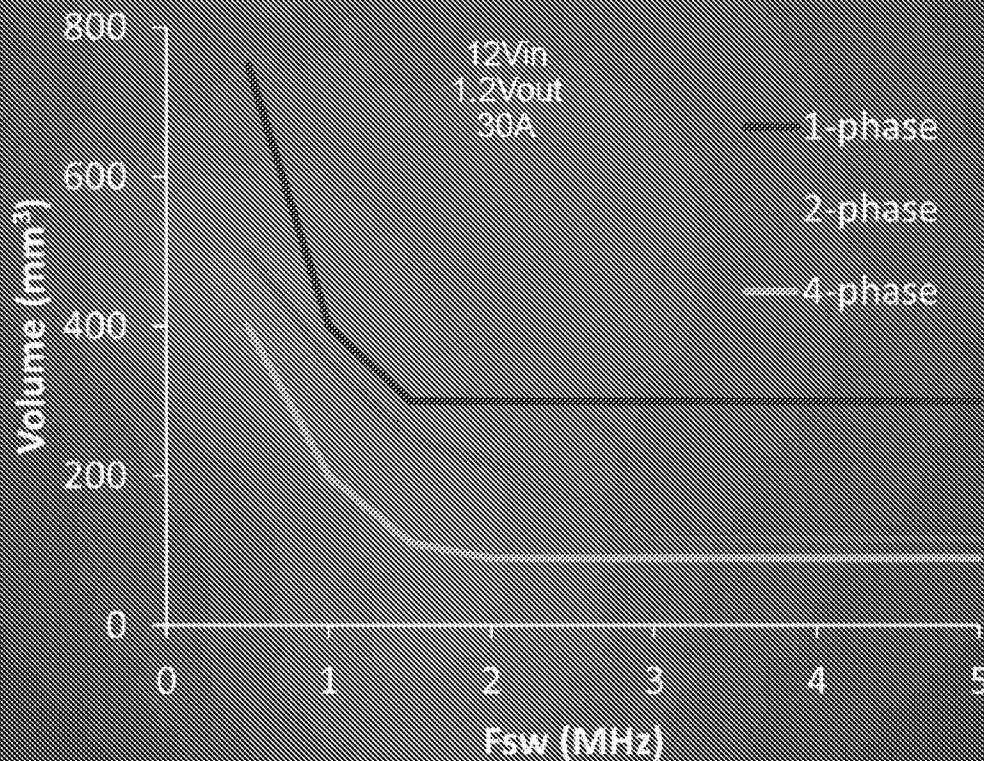
Smaller form-factor Inductors – current capability drops

Coilcraft XAL, XFL (solid lines),
Vishay IHLP (dashed lines)
families shown

Challenges to Higher Density 12Vin POL solutions

(1) FET Switching Losses → Sarda GaAs FET

(2) High current inductors are large → Multiphase (lower I/phase)

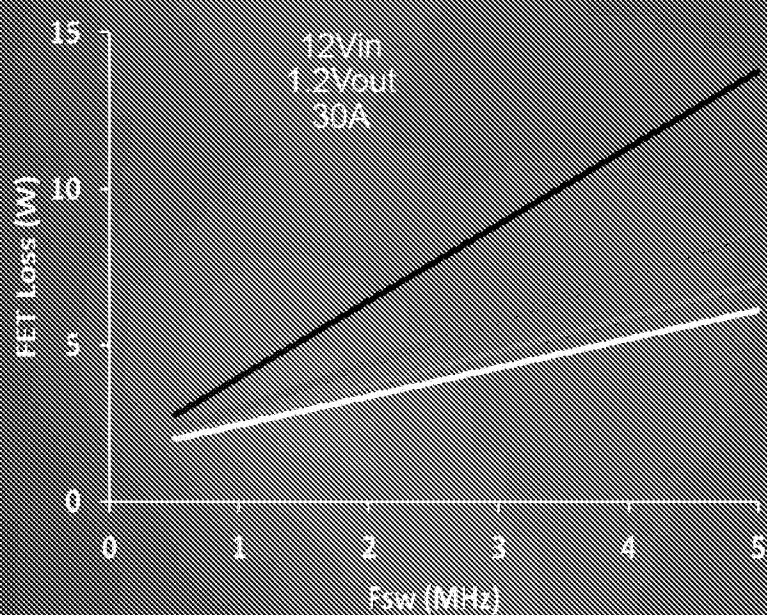


**Inductor Volume
decreases with higher
Phase count**

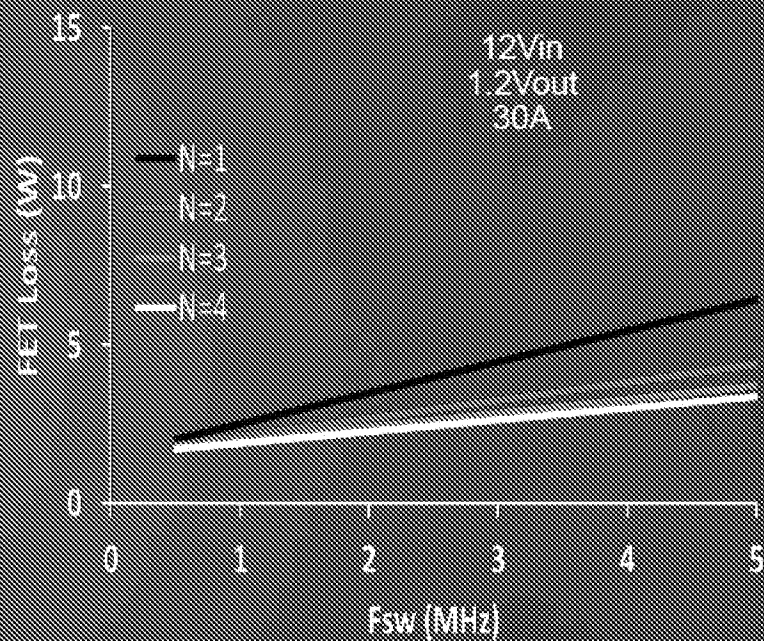
General example/trend
More than one inductor solution
choice per conditions examined

Another Multiphase Benefit

FET switching losses reduced



MOSFETs

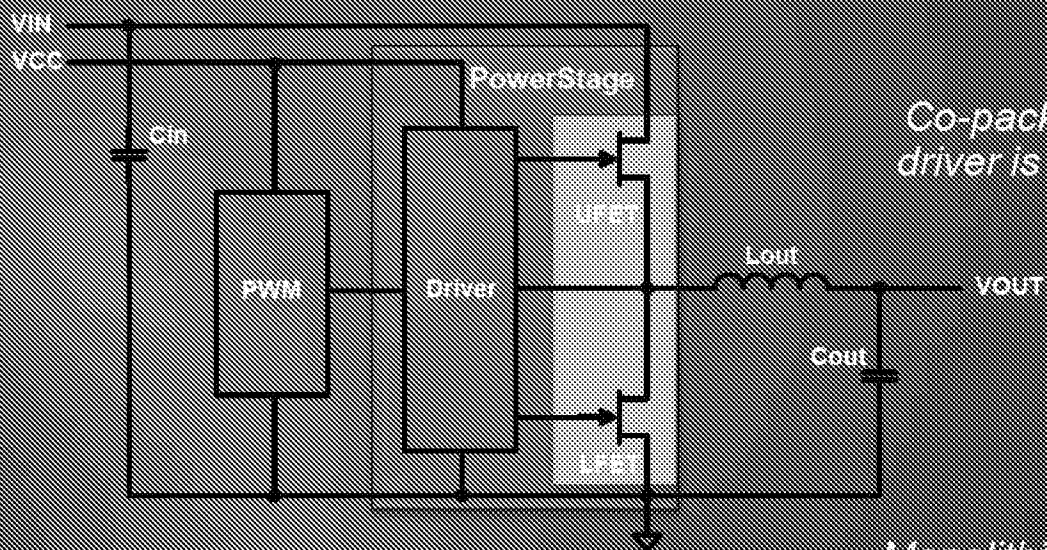


*Sarda GaAs FETs **

* In HIPS module (explained later)

Challenges to Higher Density 12Vin POL solutions

- (1) FET Switching Losses → Sarda GaAs FET
- (2) High current inductors are large → Multiphase (lower I/phase)
- (3) Parasitic Impedances

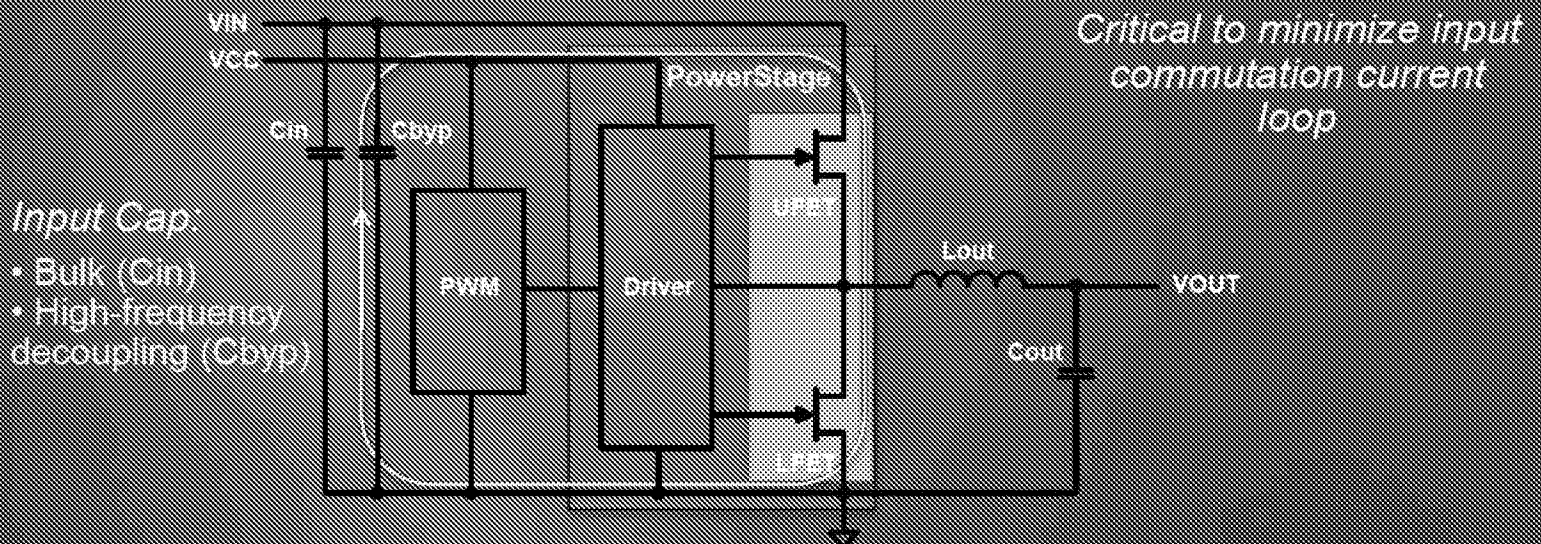


Co-packing FETs and driver is a requirement

Monolithic UFET / LFET minimizes critical common source inductance

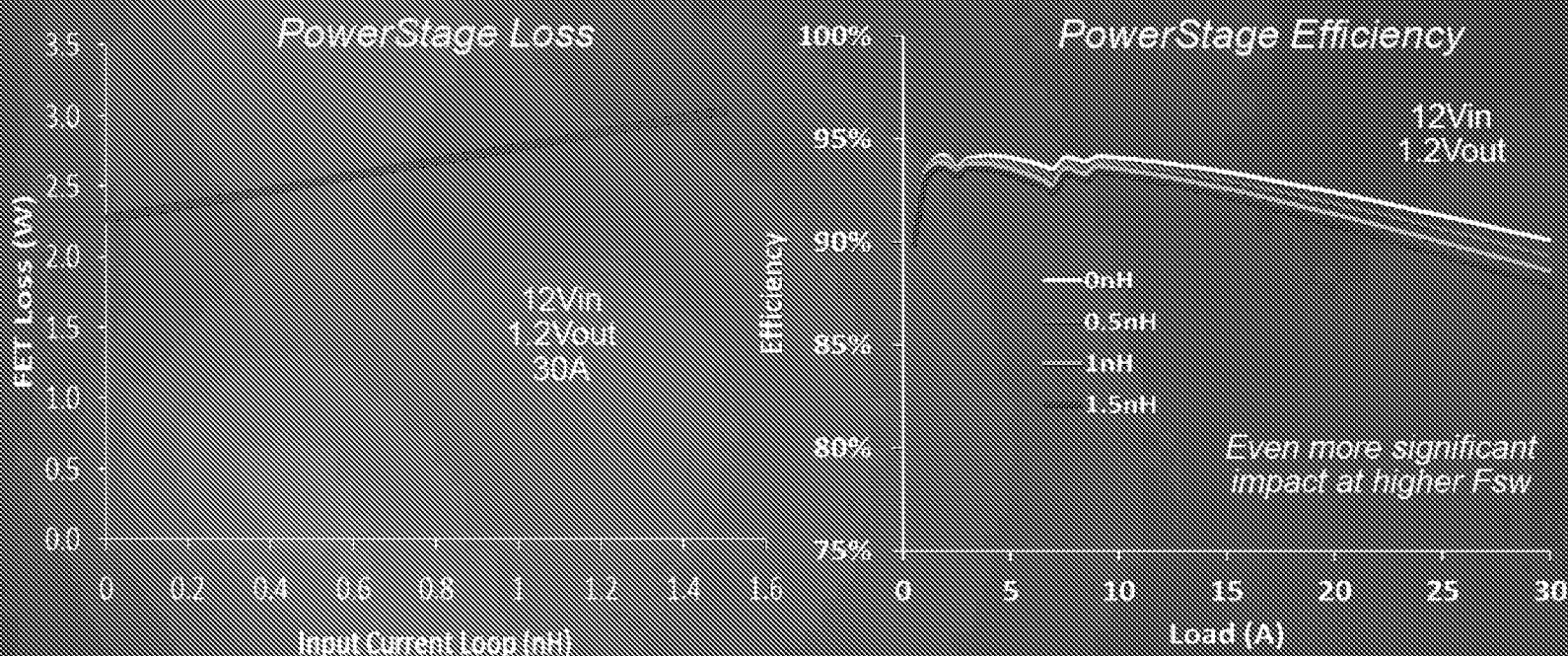
Challenges to Higher Density 12Vin POL solutions

- (1) FET Switching Losses → Sarda GaAs FET
- (2) High current inductors are large → Multiphase (lower I/phase)
- (3) Parasitic Impedances



Minimize Input Current Loop

Quantifying with Sarda Gen1 HIPS (4-phase, 2MHz)



Additional Information on impact of Input Current Loop Inductance:

- APEC 2013: "Understanding the Effect of PCB Layout on Circuit Performance in a High Frequency Gallium Nitride Based Point of Load Converter", David Reusch, et al.
- IMAPS 2013: "A Highly-Integrated GaAs-based Module for DC-DC Regulators", Greg Miller, et al.

Challenges to Higher Density 12Vin POL solutions

- (1) FET Switching Losses → Sarda GaAs FET
- (2) High current inductors are large → Multiphase (lower I/phase)
- (3) Parasitic Impedances → Integrated Power Stage

